

Faculty of Information Technology and Artificial Intelligence

Networks and Systems Administration / 10687341

Command Line Interface (CLI)
And AI agent
In Linux operating system

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1. *Introduction:*

The purpose of this practical work was to explore basic Linux commands and prepare a small AI-assisted network agent.
The agent uses a local language model through Ollama to interpret a user request, or shell gpt to run / execute Linux commands.

2. *Environment and Tools:*

Linux virtual machine running in VMware (Ubuntu).

Ollama, which runs a language model locally.

Qwen 2.5 1.5B, the lightweight model selected for the agent.

ShellGPT 1.5.1

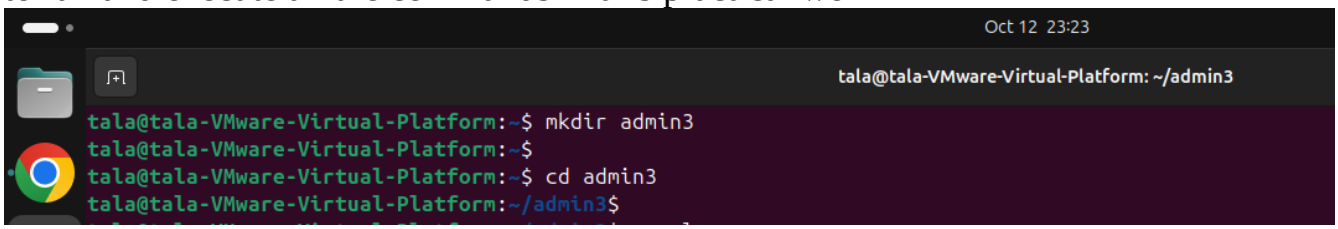
Python virtual environment, Python 3

3. *Procedure*:

3.1 executing the commands in lecture 3 (intro to linux) on ubuntu terminal in the VMware:

3.1.0

First, created a new directory named admin3 with command **mkdir admin3** and entered it with cd **admin3**
to run and execute all the commands in this practical work

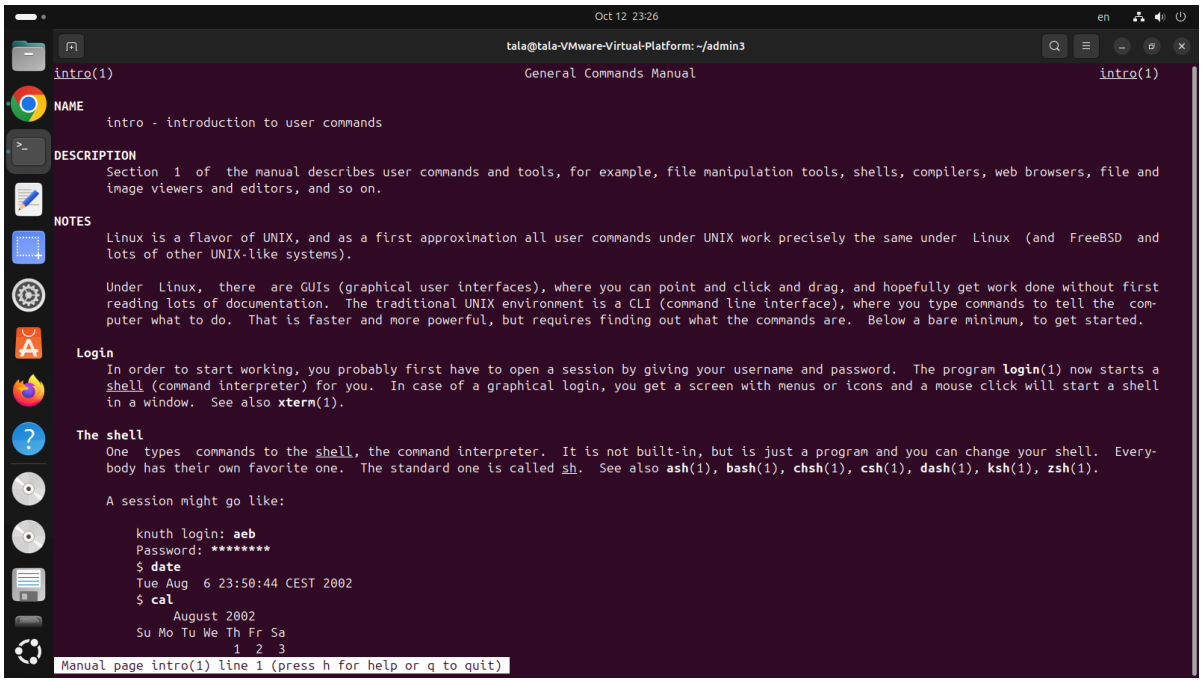


3.1.1 man commands:

The man commands (Manual) used to view the reference manuals for commands.

man n intro

view the introduction to the user commands manual section, displays general information about the manual and user commands in Linux.

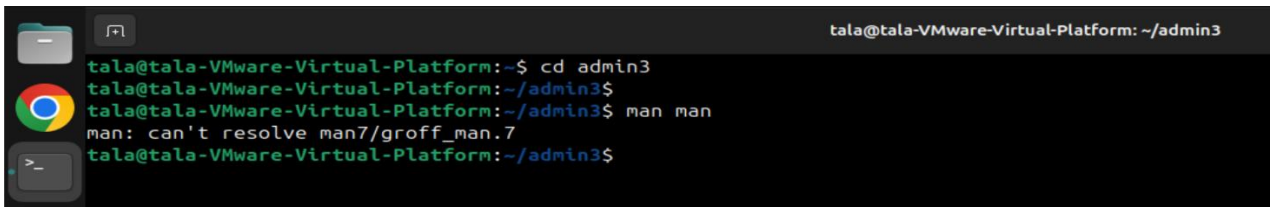


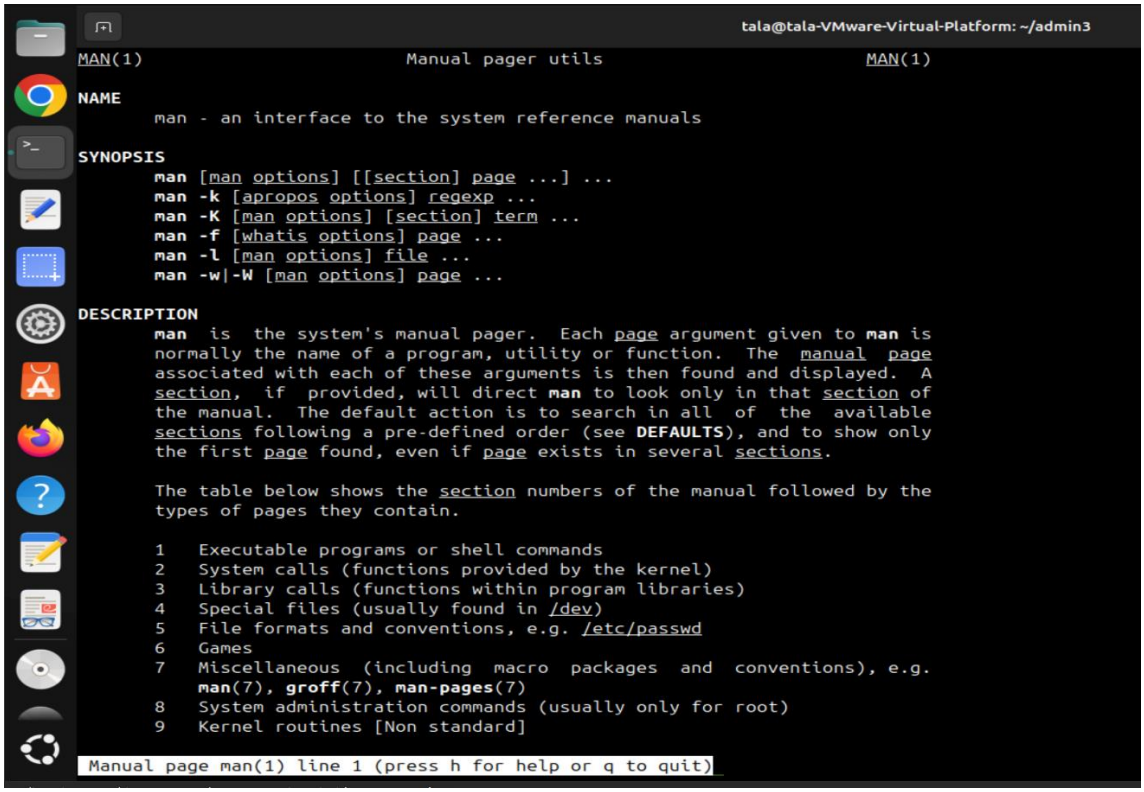
man n

View the manual page for the n.

man man

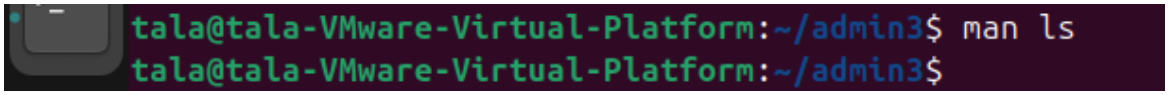
Opens the manual page for the man itself, it’s the instruction manual that explains how to use the manual command.

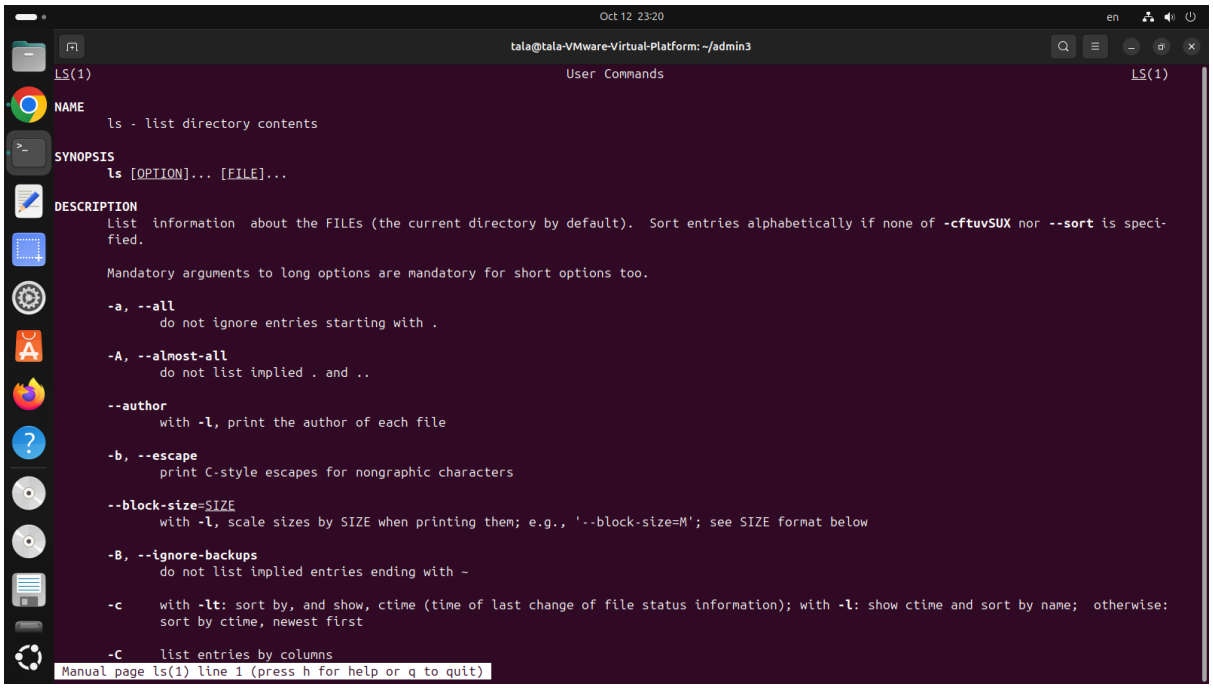




man ls

Opens the manual page for the ls (list directory contents)

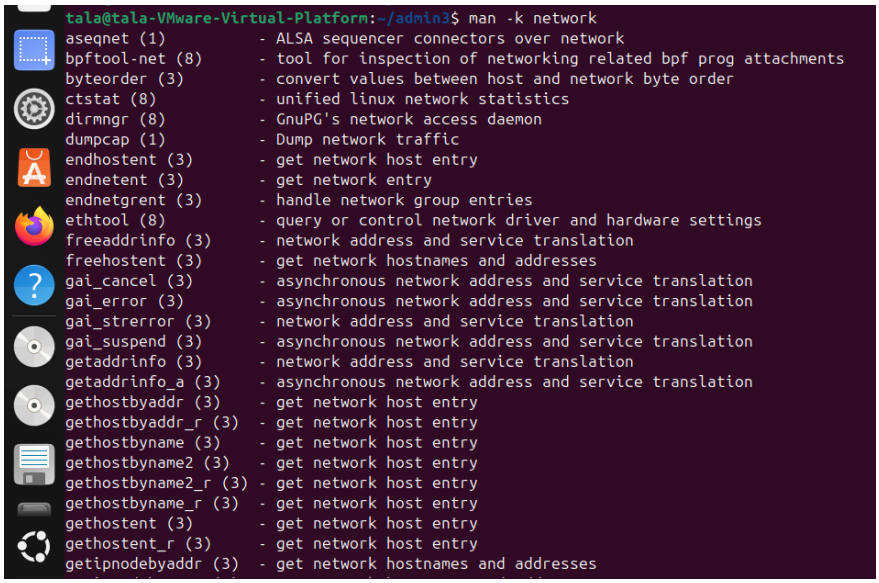




man –k string

Search the short descriptions and manual page titles for a keyword (apropos search)

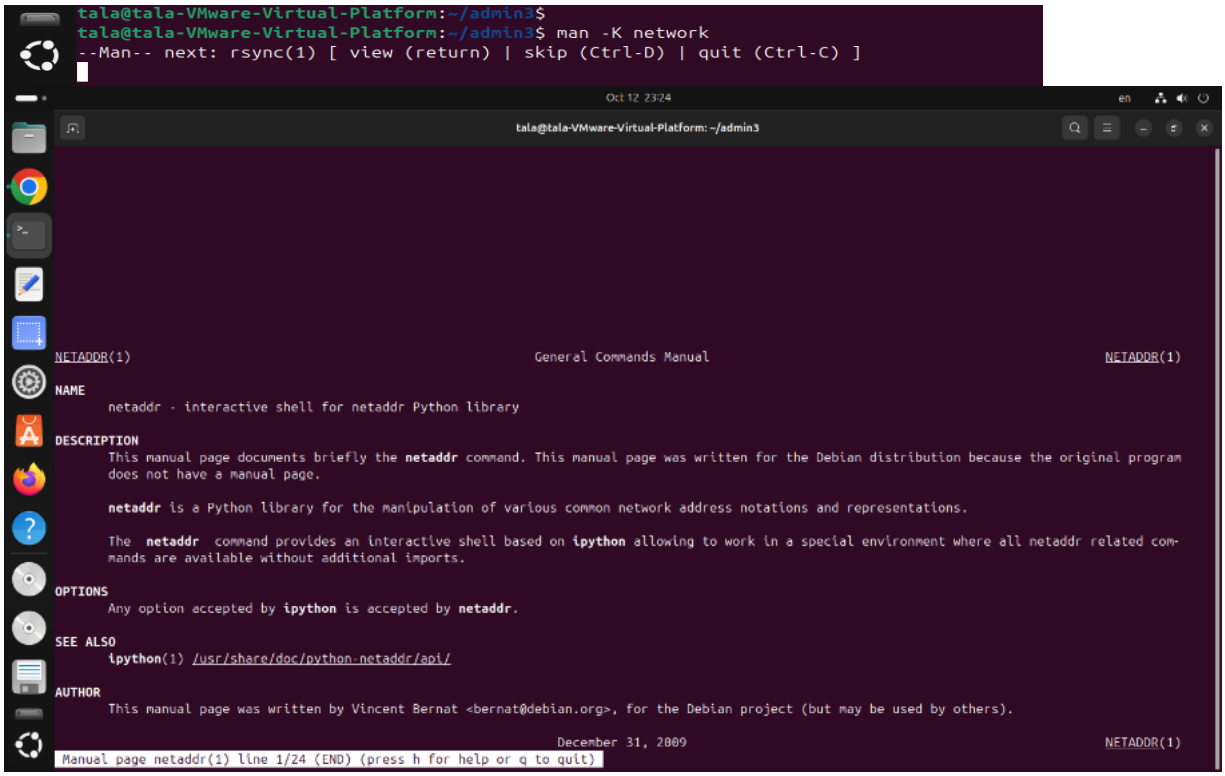
man –k network



man –K string

Search for string inside the full text of manual pages.

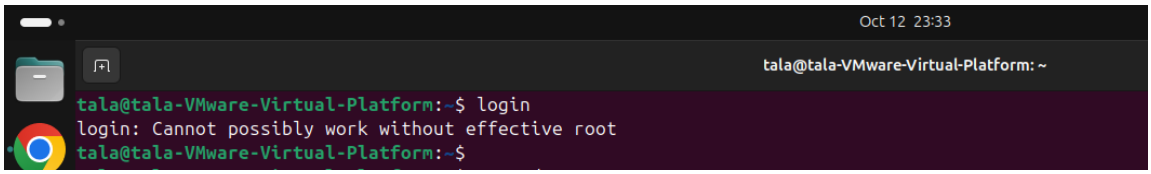
man –K network



3.1.2 Account Related Commands:

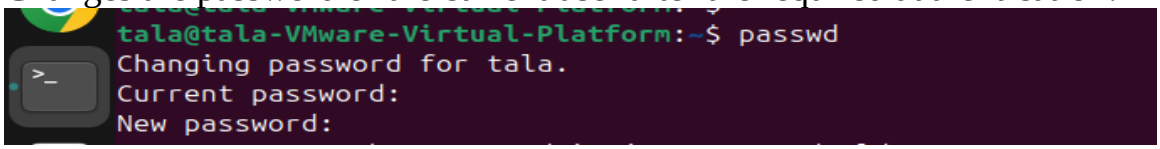
login

Starts a new session.



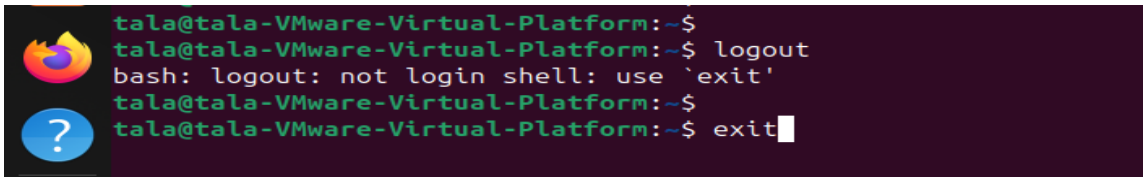
passwd

Changes the password of the current user after the required authentication.



logout / exit

Closes / ends the current shell session.

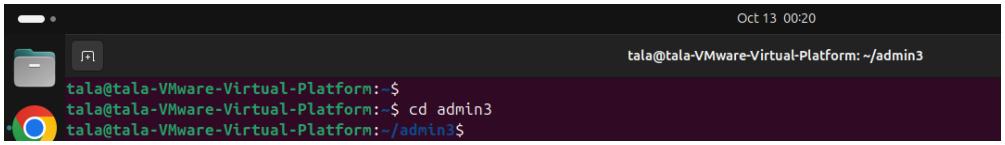


3.1.3 File/Directory Commands:

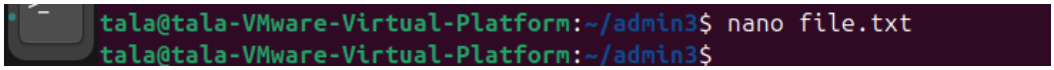
- Files management

Executed (cp, mv, rm, cat, more, less) commands on some files in the directory admin3 that was created previously.

Entered the directory admin3 by command **cd admin3**

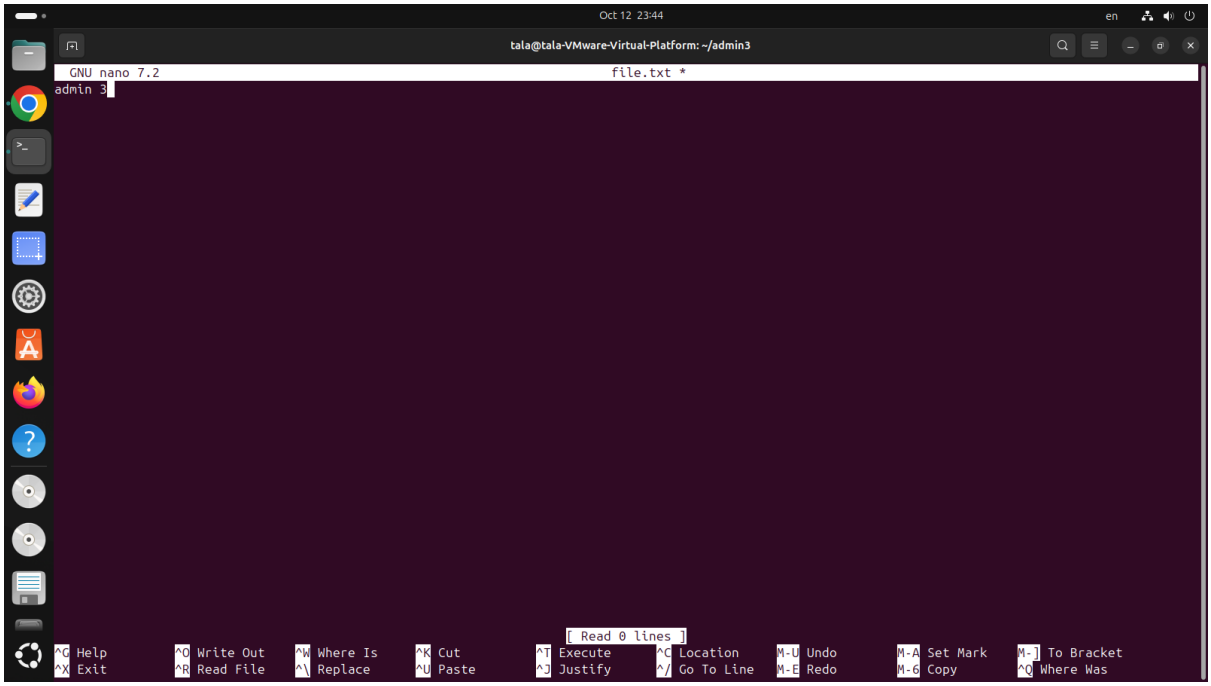


Made a file to run the commands on it, the command used, **nano file.txt**



Write in the file.txt (admin 3)

Then close the file by press (control+x), that will display a question if you want to save the file or not, saved the file and confirmed its name (file.txt)

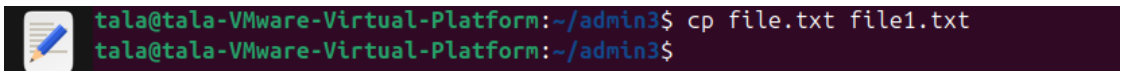


cp file1 file2|dir1

Copy file1 to file2 or into directory dir1

cp file.txt file1.txt

Copies file.txt to a new file named file1.txt

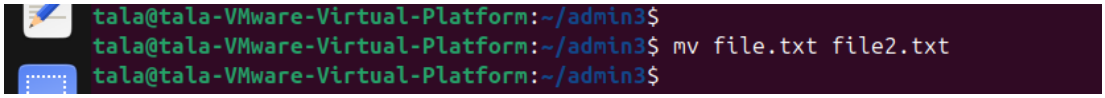


mv file1 file2|dir1

Moves file1 to file2 or into directory dir1
Renames file1 to file2 if both in same directory

mv file.txt file2.txt

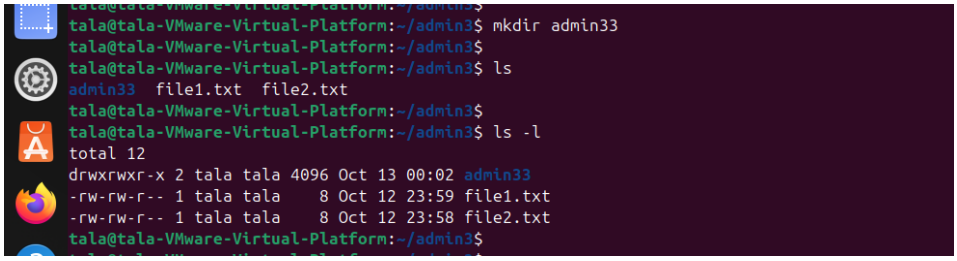
Renames file.txt to file2.txt, if both in same directory



The mv command can also move files to another directory.

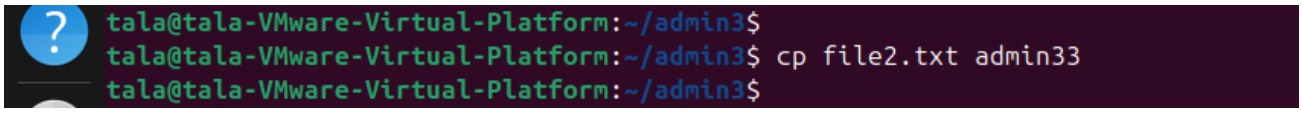
Created a new directory inside the admin3 directory, named admin33,
mkdir admin33

ls to make ensure that it was created, **ls -l** to view more details.



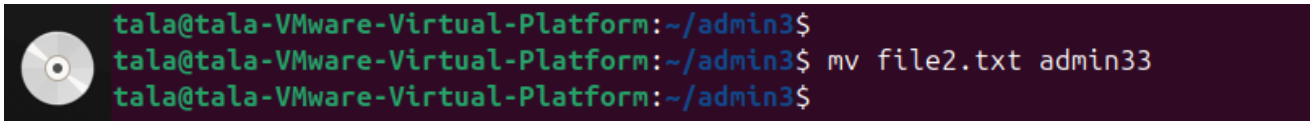
cp file2.txt admin33

Copies file2.txt to the new directory that just created (admin33).



mv file2.txt admin33

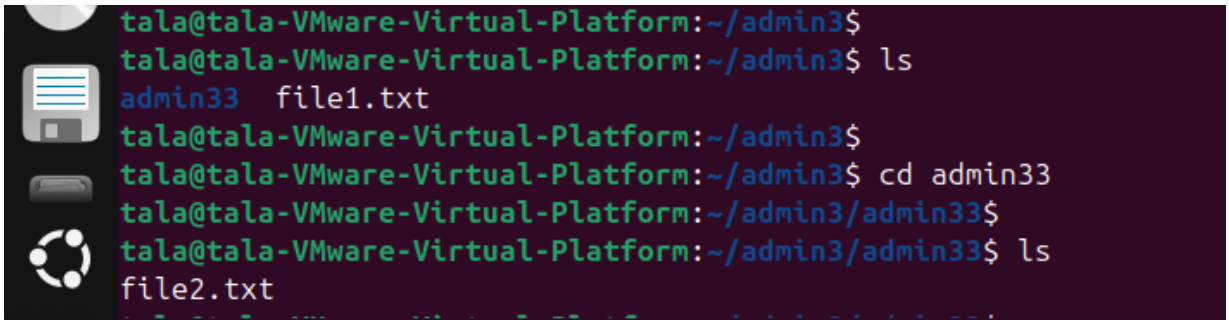
Moves file2.txt to the new directory admin33



ls to view that it was moved from admin3 directory

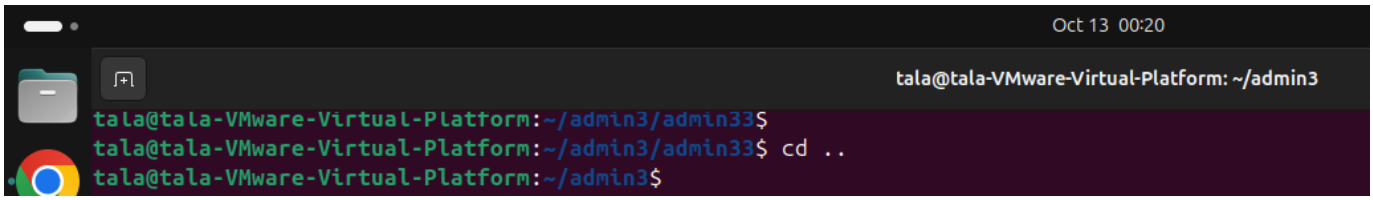
cd to enter the new directory admin33

ls inside the new directory admin33 to view that it contains the file2.txt that was moved from admin3 directory.



cd ..

Moves to the parent directory (admin3)

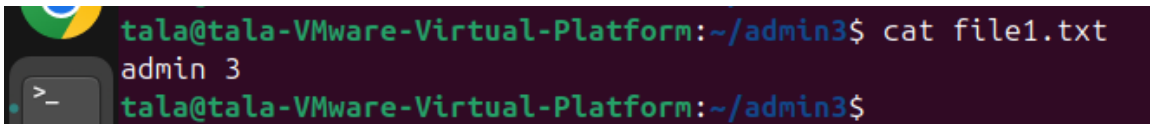


Viewing files:

Run the commands (cat, more, less) on the files were created previously.

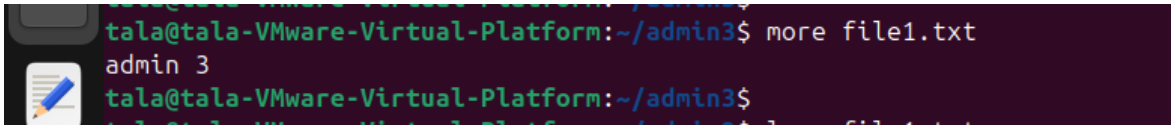
cat file1.txt

Displays the entire contents of the file1.txt to the screen.



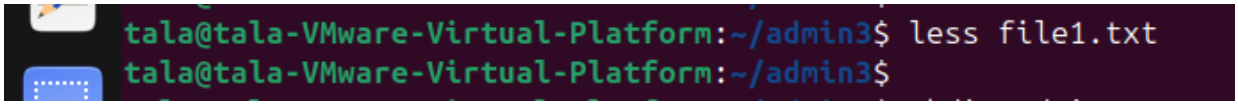
more file1.txt

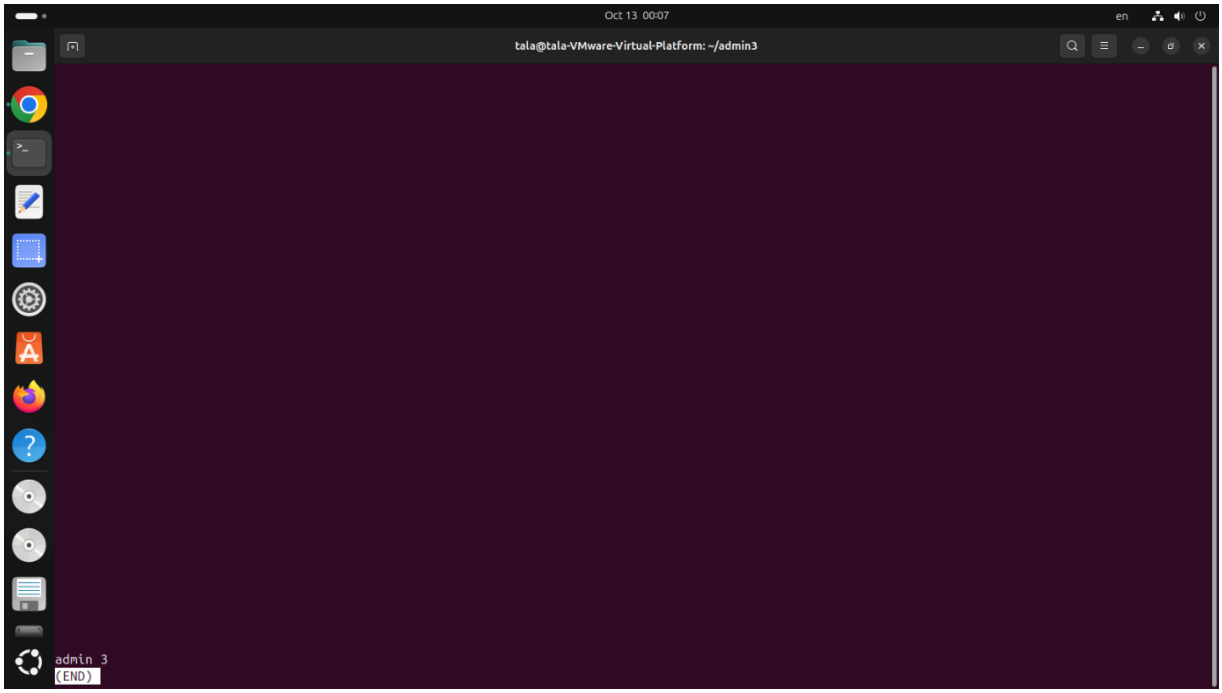
Displayed the contents of file1.txt, one screen at a time. Press q to quit more.



less file1.txt

Same as more but more powerful. Allows to navigate through file1.txt, including moving forward and backward. Press q to quit less.





- Directories management

Executed (mkdir, ls, pwd, cp, cd, rm/rmdir, mv) commands on directories that will be created in the directory admin3.

mkdir dir1

Creates directory dir1

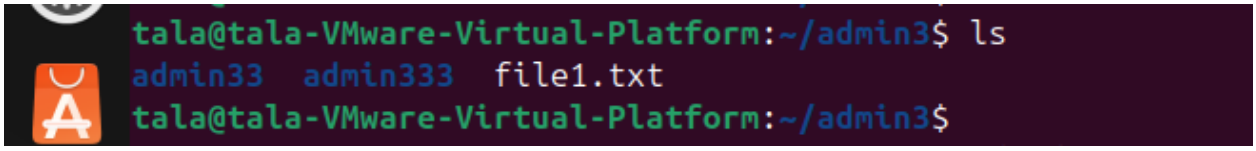
mkdir admin333

Created a new directory (admin333) in admin3 directory.



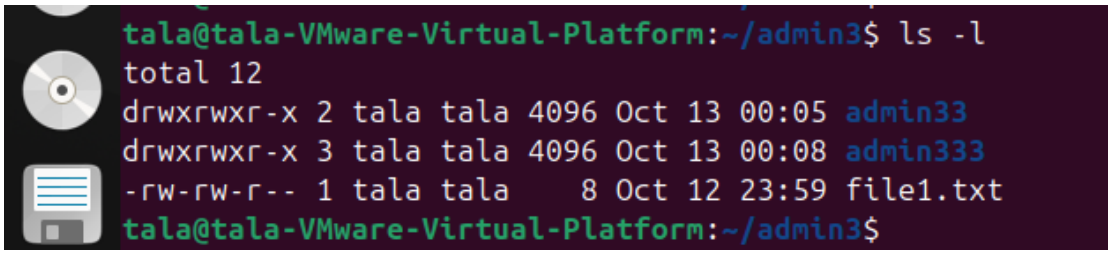
ls

Lists contents (files + directories) in the current directory



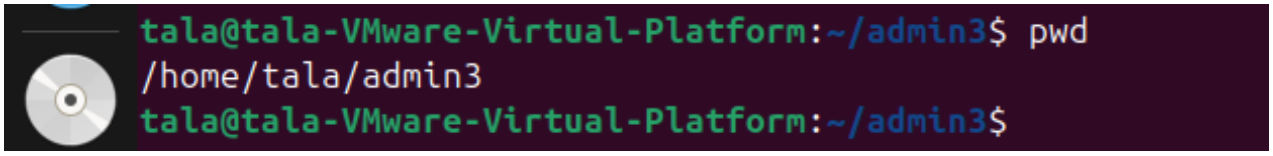
ls [-la] [file/dir list]

Displays all entries, including hidden files, with detailed information.



pwd

Displays the full path of the current working directory.

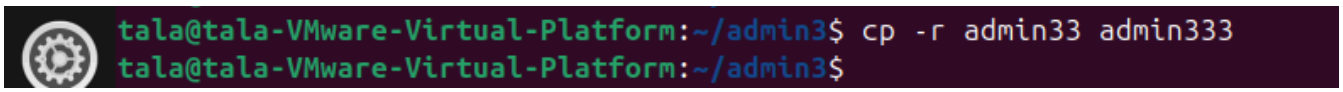


cp –r[p] dir1 dir2

Copy directory dir1 to dir2

cp –r admin33 admin333

To copy a directory and its contents, the -r option enables recursive copying.



cd dir1

Changes the current directory to dir1, if that directory exists in the current location.

cd admin333

Entered the directory admin333



to check its content that just was copied (admin33 directory), **ls**



cd ..

Exited the new directory admin333, moved / backed to the parent directory (admin3)

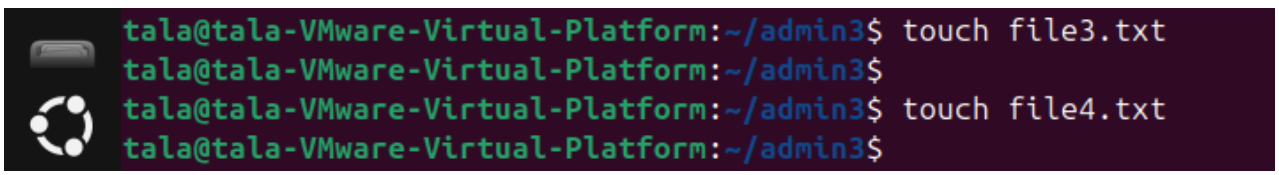


Removing files and directories:

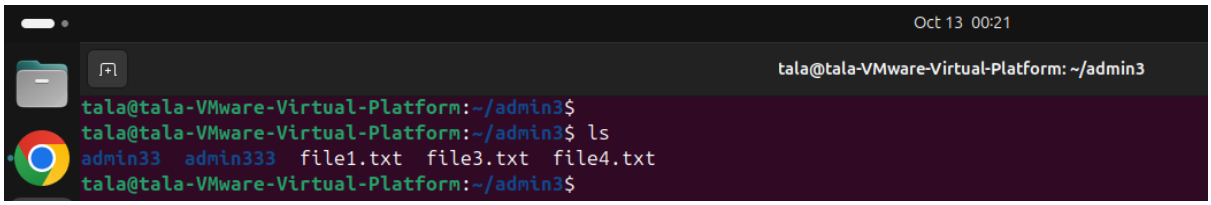
Created new files to execute the removing commands (rm)

touch file3.txt

touch file4.txt



ls to display the contents of the admin3 directory

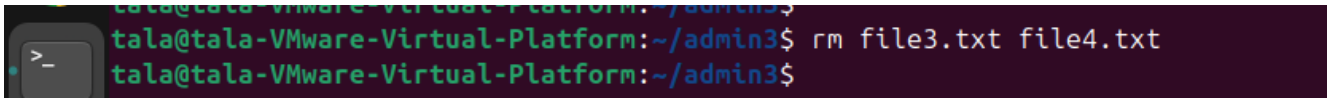


rm file1 file2 ...

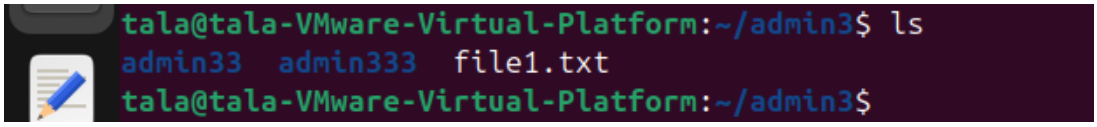
Removes list of files.

rm file3.txt file4.txt

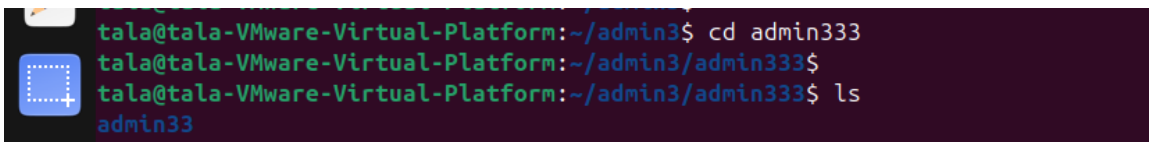
Removes the 2 files were just created.



ls to display the remaining contents.



Entered the admin333 directory, **cd admin333**
Display the contents, **ls**

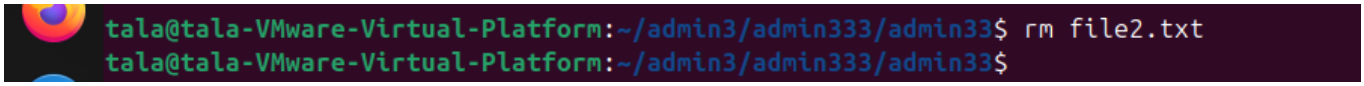


Entered the admin33 directory, **cd admin33**
Display the contents, **ls**

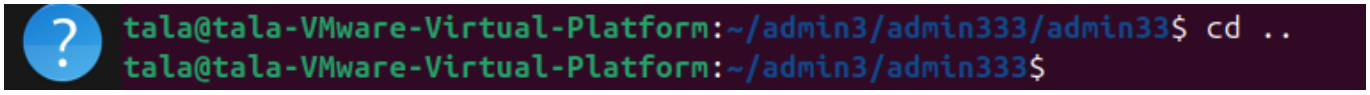


rm file2.txt

Removes only a single file, file2.txt



Exited from admin33, returned back to admin333, **cd ..**

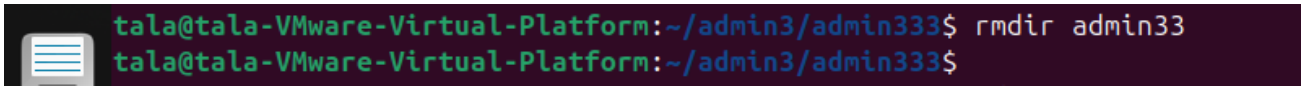


rmdir dir1

Removes dir1 only (if it is empty).

rmdir admin33

Removes the empty admin33 directory, that just removed the file2.txt from.



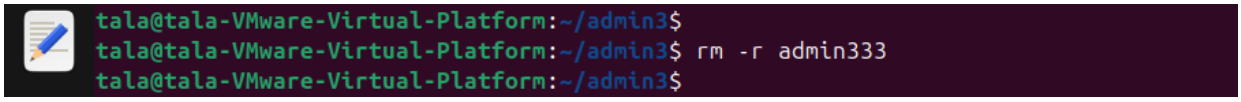
Exited from admin333, returned back to admin3, **cd ..**

Display the contents, **ls**

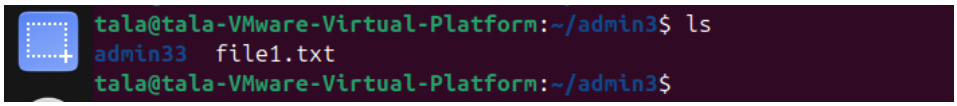


rm -r dir1

Removes dir1 and its contents (all subdirectories/files), it’s very dangerous, so it must be used carefully.

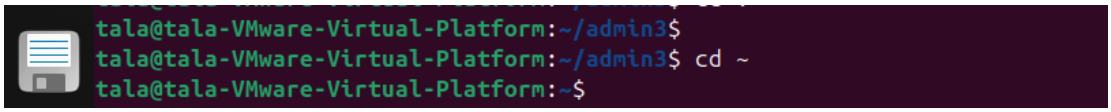


To check, **ls**



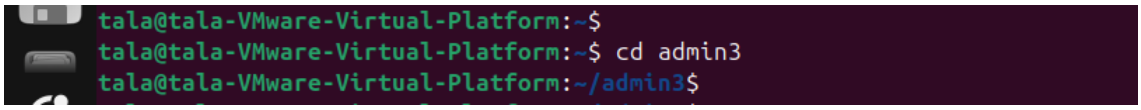
cd ~

Returns to the user's home directory.



cd admin3

Enters admin3 directury



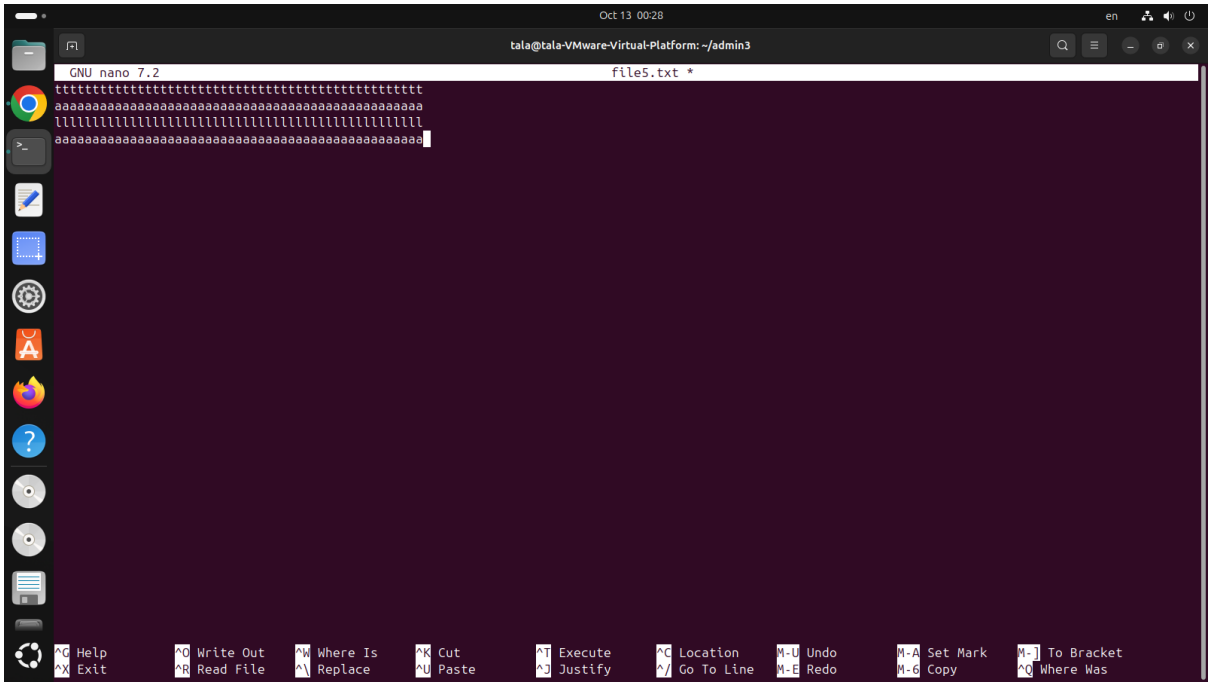
3.1.4 Filter Commands:

Executed (cat, more, less, head, tail, wc, sort, uniq, cut, paste, grep, tr) commands on some files in the directory admin3.

Created a new file named file5.txt, **nano file5.txt**

And wrote in it, then saved it.

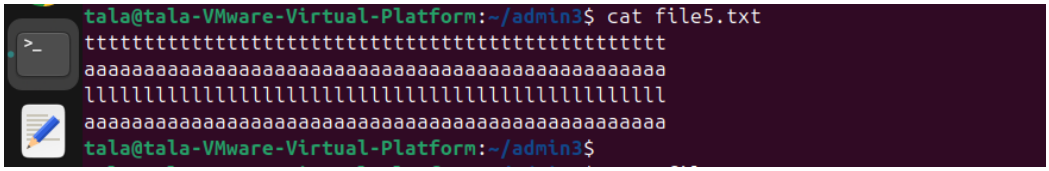




cat file

Displays entire file content on the screen

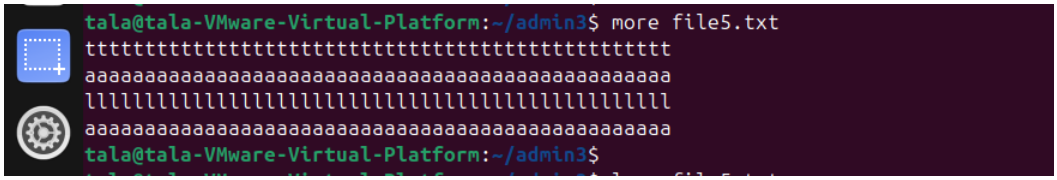
cat file5.txt



more file

Displays file one screen at a time

more file5.txt

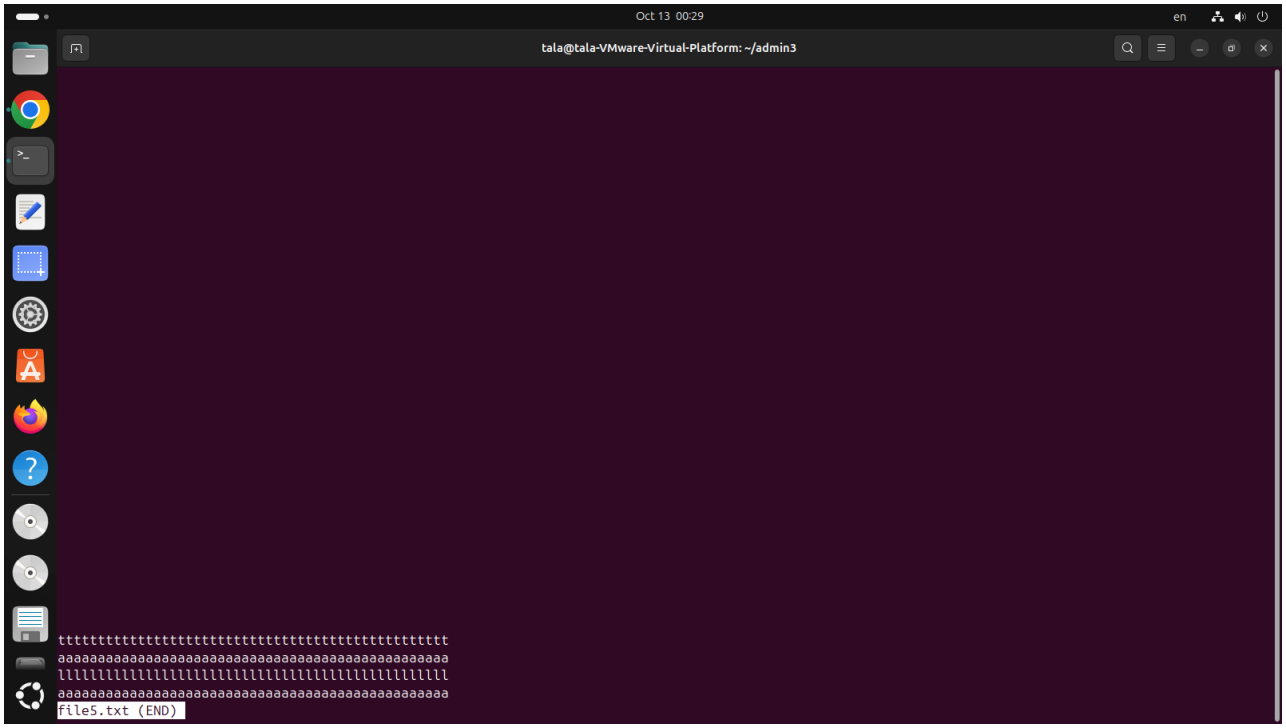


less file

Displays file, more powerful. Allows to navigate through the file.

less file5.txt

```
tala@tala-VMware-Virtual-Platform:~/admin3$ less file5.txt
tala@tala-VMware-Virtual-Platform:~/admin3$
```



head -# file

Displays the first # lines of file.

head -n 1 file5.txt

Displays the first line of file5.txt

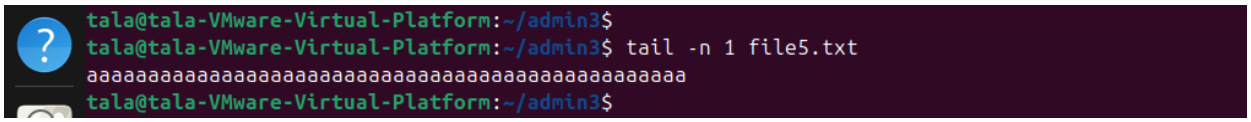
```
tala@tala-VMware-Virtual-Platform:~/admin3$
tala@tala-VMware-Virtual-Platform:~/admin3$ head -n 1 file5.txt
tttttttttttttttttttttttttttttttttttttttttttttttttttttttttt
tala@tala-VMware-Virtual-Platform:~/admin3$
```

tail -# file

Displays the last # lines of file

tail -n 1 file5.txt

Displays the last line of file5.txt



wc [-cwl] file

Counts number of characters, words, or lines in a file and displays it.

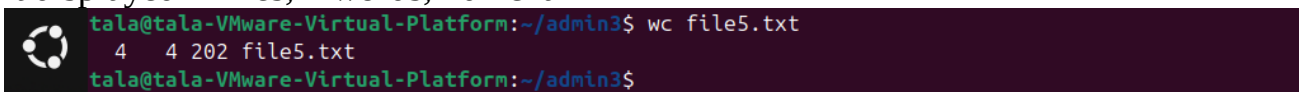
-l: Counts lines.

-w: Counts words.

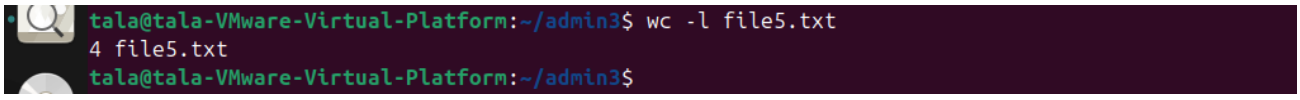
-c: Counts bytes.

wc file5.txt

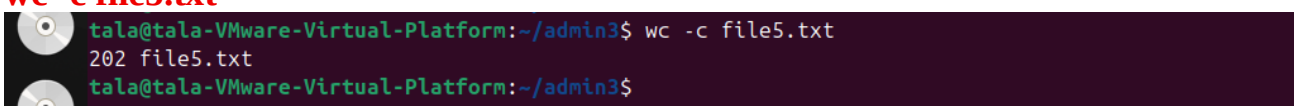
it displayed 4 lines, 4 words, 202 char



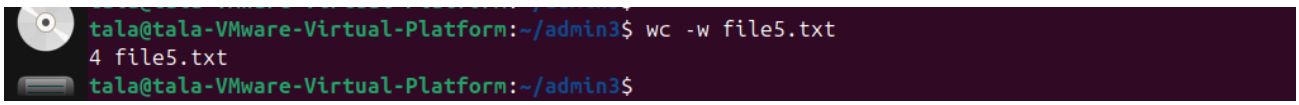
wc -l file5.txt



wc -c file5.txt



wc -w file5.txt



Sorting:

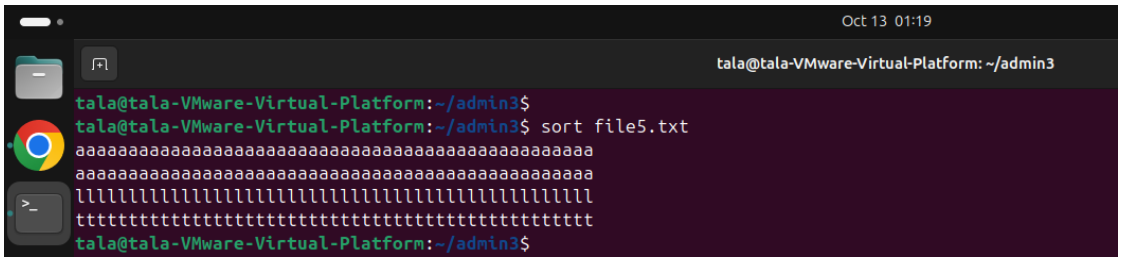
Lists the contents of a file based on order, sort by field.

sort file

Sorts file alphabetically by line

sort file5.txt

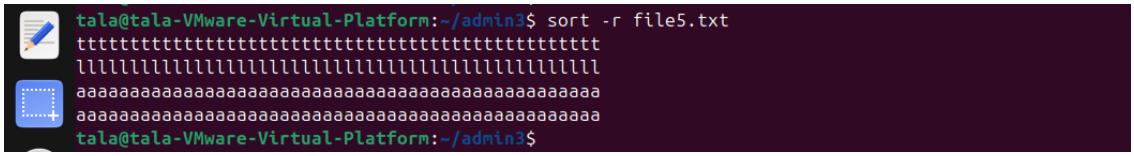
Displays the file lines in sorted order.



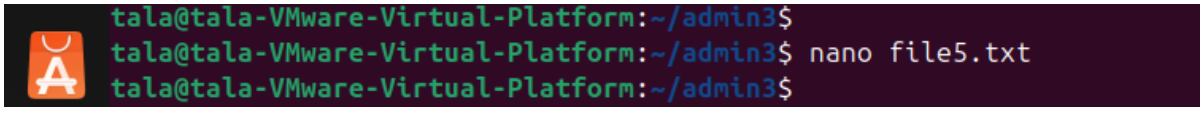
sort -r file

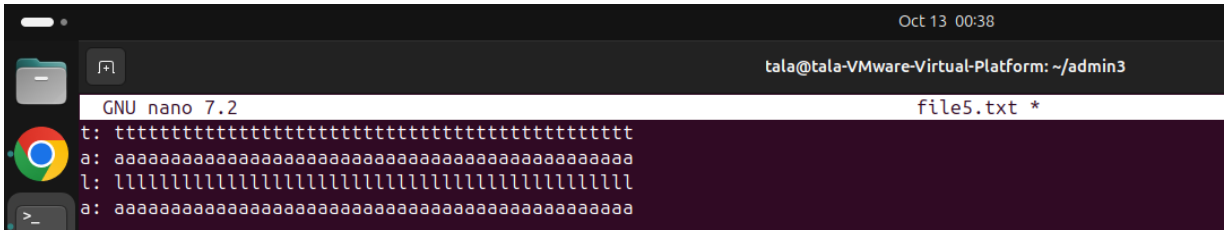
Sorts file in reverse order by line, displays it

sort -r file5.txt



Made some changes on file5.txt, **nano file5.txt**





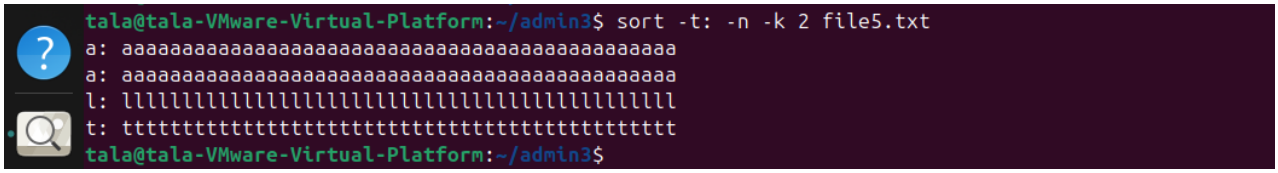
sort –t: -n –k # file

Sorts file based on the field number (-k)

in numeric order (-n)

with fields separated by a char after (-t)

sort –t: -n –k 2 file5.txt


tala@tala-VMware-Virtual-Platform:~/admin3$ sort -t: -n -k 2 file5.txt
a: aaaa

Extracting info:

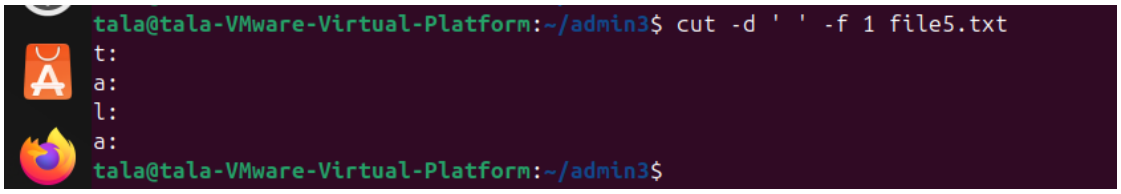
cut – Get fields

cut –f# [-d%] file

Displays # fields separated by % in file

cut –d ‘ ’ -f 1 file5.txt

Uses the space ( ) as a delimiter and displays the first field of each line in file5.txt

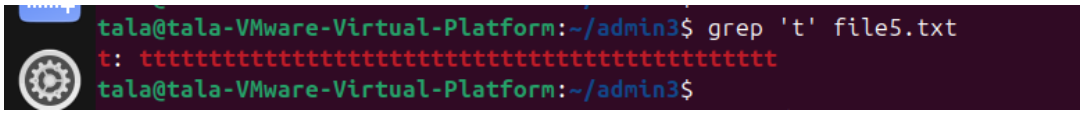


grep search-string file

To search text and displays all lines with search-string in file

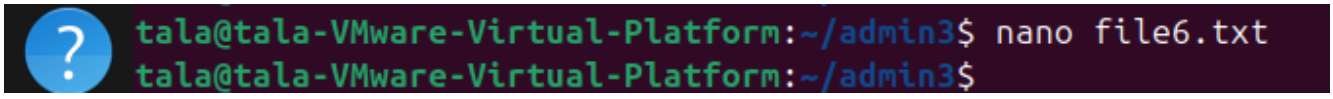
grep ‘t’ file5.txt

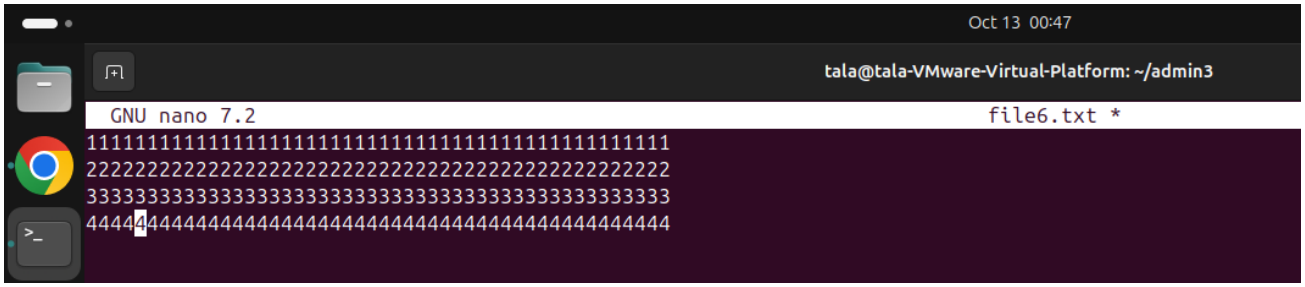
Searches for lines containing ‘t’ and displays matching lines.



Changing file contents:

First, created a new file named file6.txt, with 4 lines, **nano file6.txt**



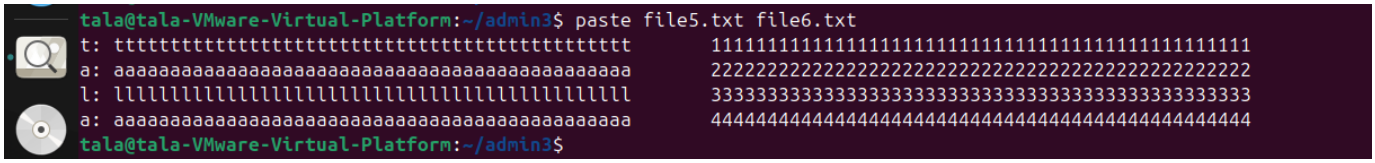


paste file1 file2

Merge contents of file1 and file2 line by line

paste file5.txt file6.txt

Combines corresponding lines from two files side by side, then displays the output

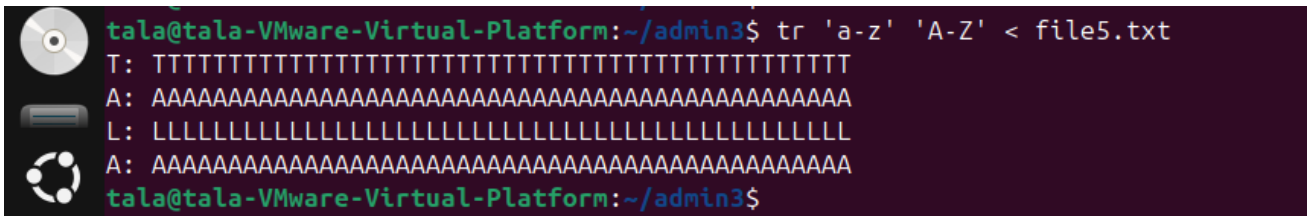


tr c1 c2 < file

Changes all occurrences of character c1 to c2 in file

tr 'a-z' 'A-Z' < file5.txt

Replaces text, Converts lowercase letters to uppercase in the displayed output. The original file is not modified by this command.

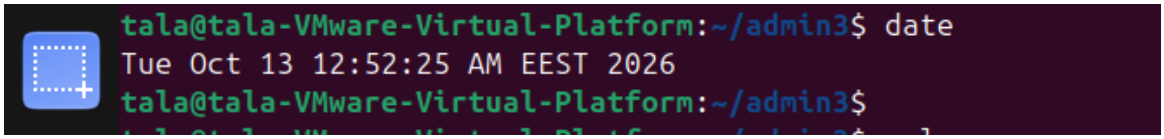


3.1.5 Misc. Commands:

System Information and Output Commands

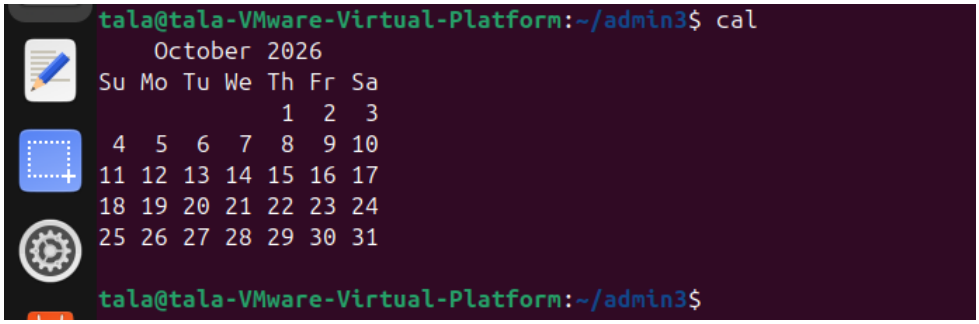
date

Displays the current system date and time.



cal

Displays a calendar for the current month.

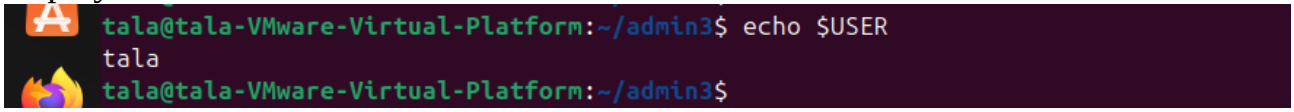


echo

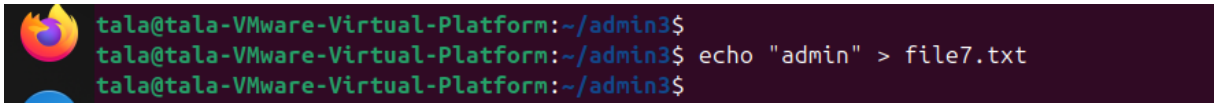
Display strings & shell variables

echo $USER

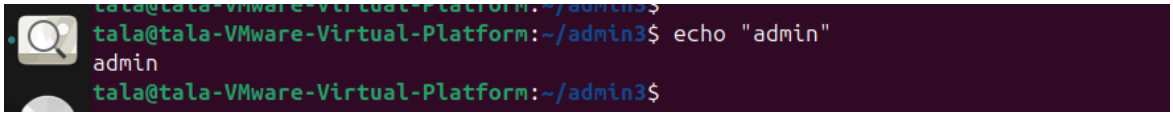
displays the name of the current user



echo “admin” > file7.txt



echo “admin”



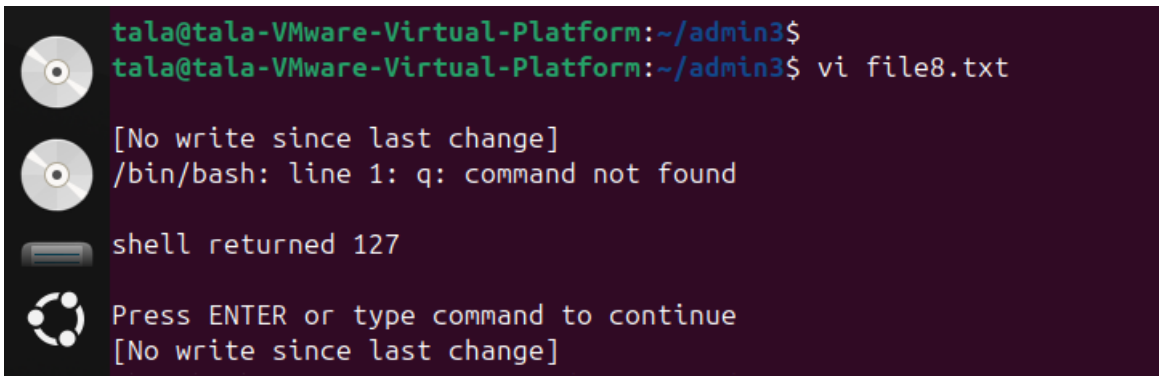
3.1.6 Visual Editor (**vi**):

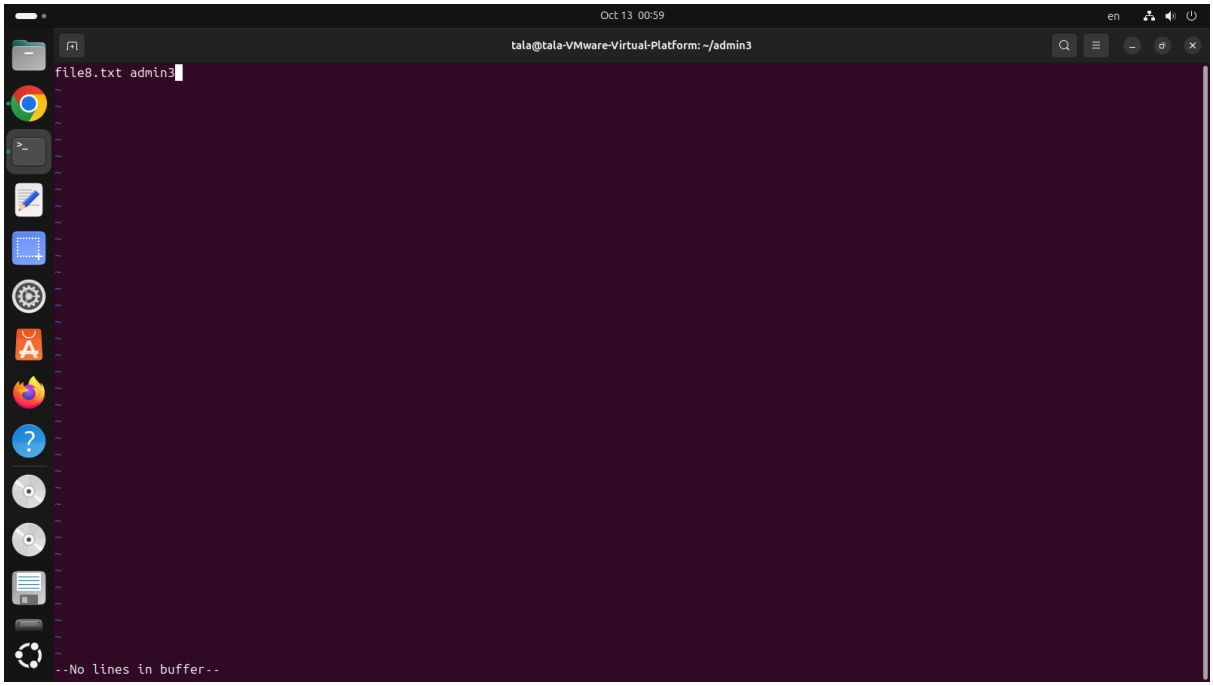
Opens a file in the editor or creates it if it does not exist.

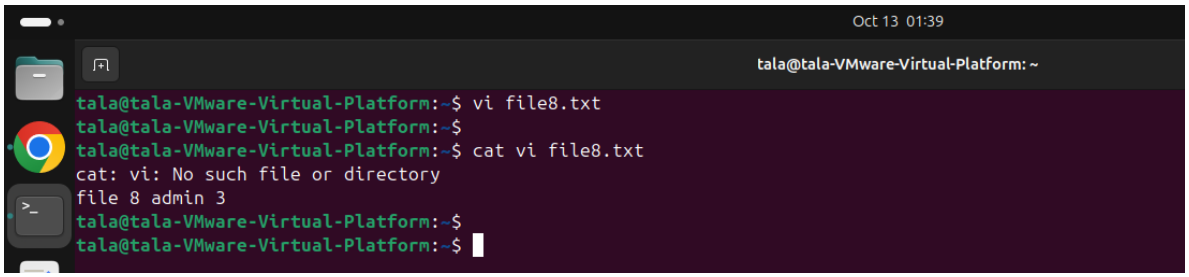
Basic operations:

- Press (i) to enter Insert mode.
- Press (Esc) to return to Command mode.
- Type (:w) to save the file.
- Type (:q) to quit.
- Type (:wq) to save and quit.
- Type (:q!) to quit without saving changes.

vi file8.txt







3.2 Using AI agent on ubuntu terminal in the VMware to execute commands by itself:

prepare a small AI-assisted agent to explore how AI can interact with a Linux operating system, generate commands based on natural-language requests, and execute them to perform system administration tasks.

The agent uses 2 implemented by two methods:

The first one is a local language model through Ollama to interpret a user request, through Python.

The second one is by using ShellGPT, existing command-generation interface.

3.2.1 Method One: Building a Custom AI Agent Using Python and Ollama:

3.2.1.1 Tools Used

- Ollama
- Qwen 2.5:1.5B language model
- Python virtual environment, Python 3

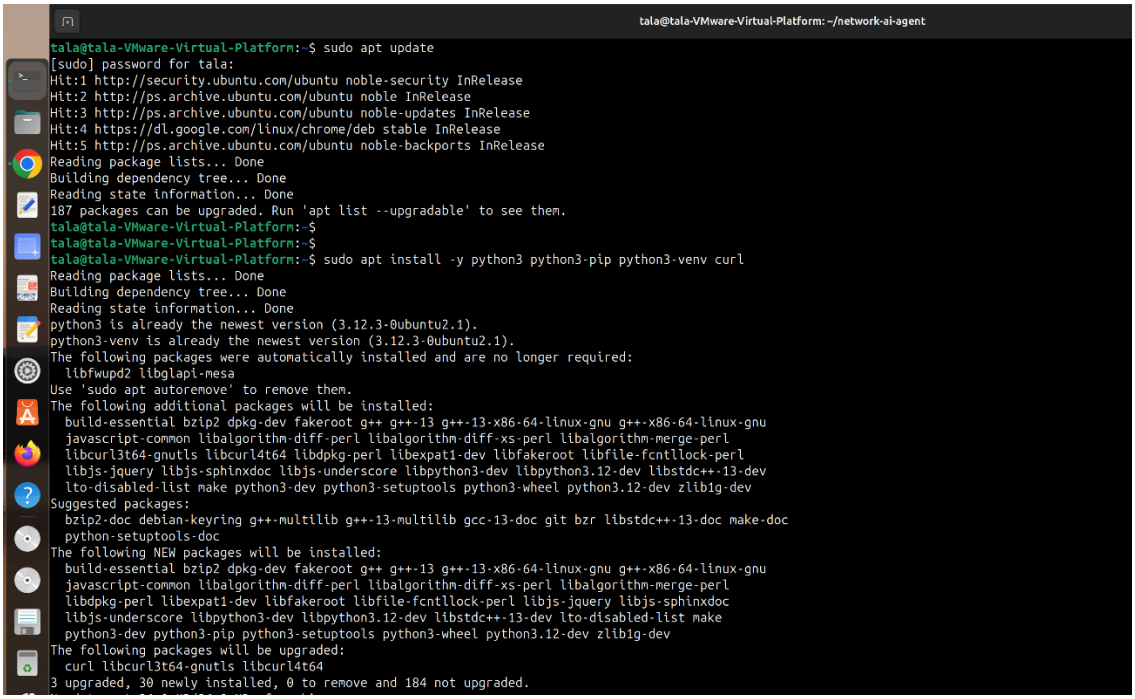
3.2.1.2 Installation and Configuration:

Installed The required packages

sudo apt update

sudo apt install -y python3 python3-pip python3-venv curl

The packages provide Python, a virtual environment, and the communication tools.



Installed ollama and and it started as a service.

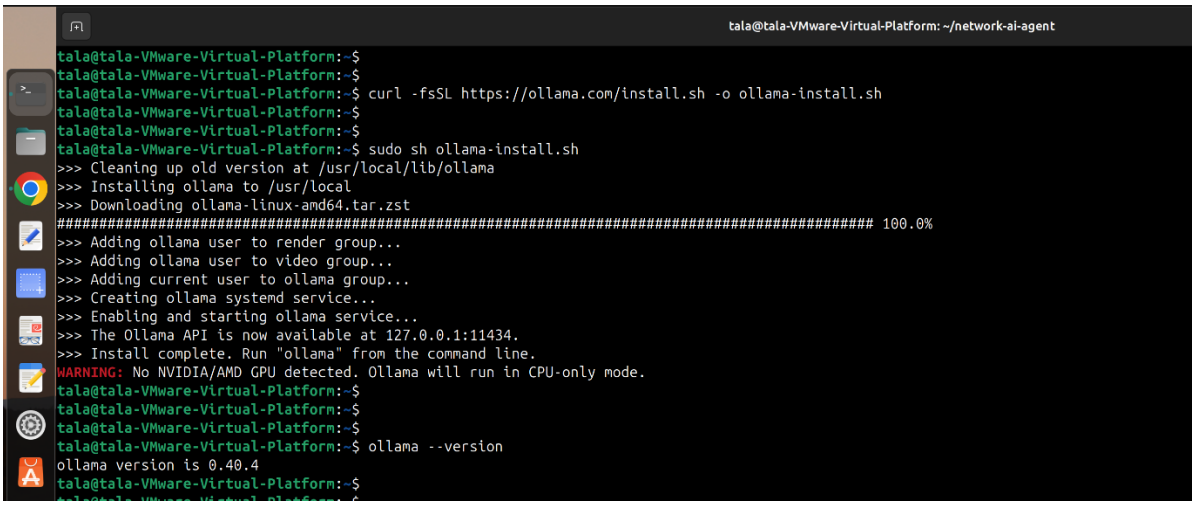
curl -fsSL https://ollama.com/install.sh -o ollama-install.sh

Run the installation file

sudo sh ollama-install.sh

confirm the installation

ollama --version



Started ollama

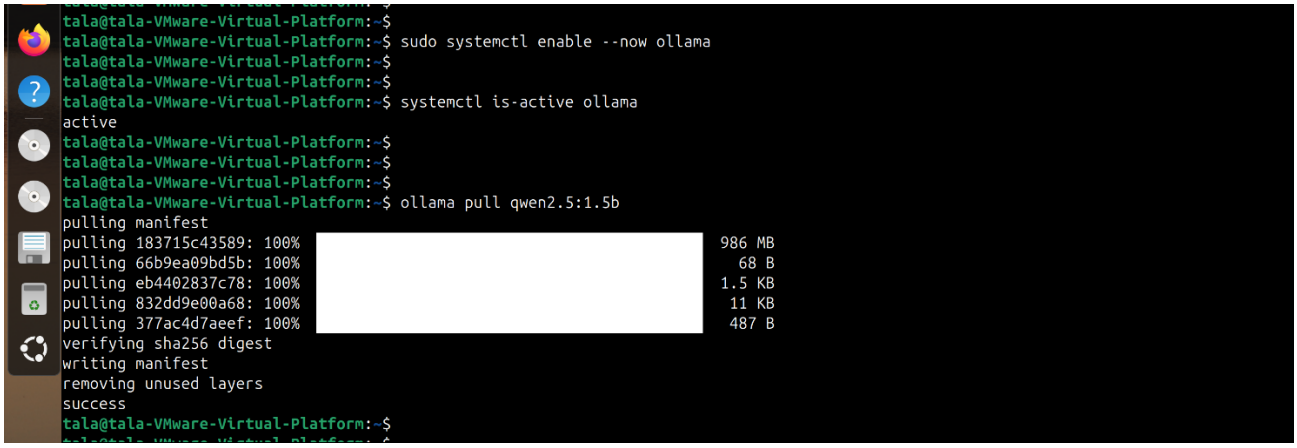
sudo systemctl enable --now ollama

Checked its status

systemctl is-active ollama

Downloaded The AI language model

ollama pull qwen2.5:1.5b



Checked that the model was installed

ollama list

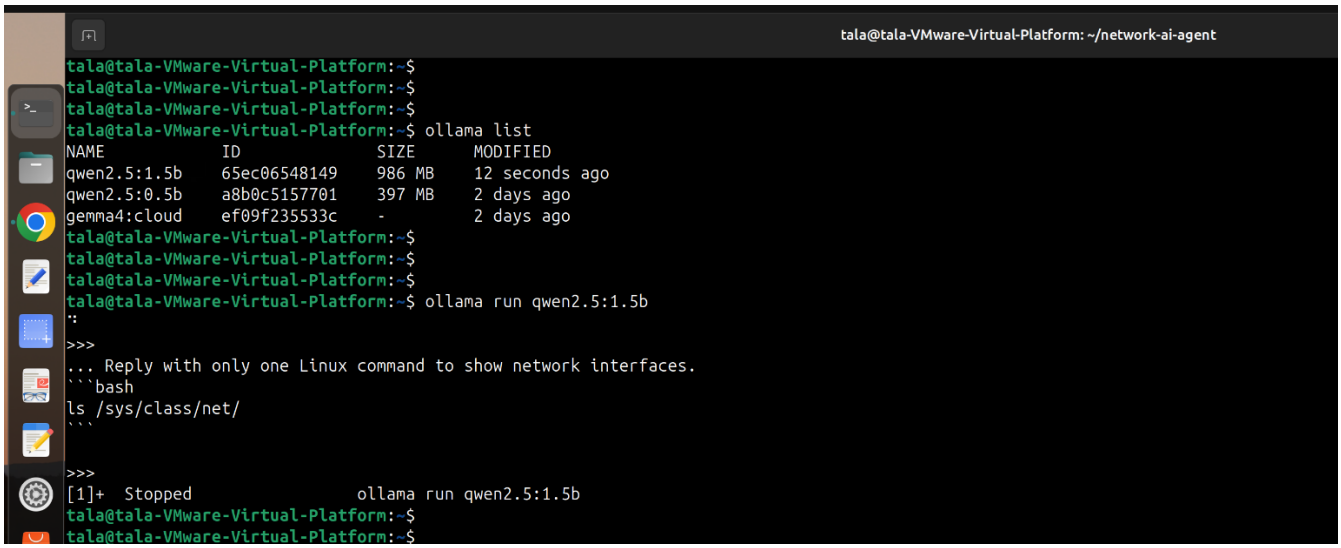
executed it,

ollama run qwen2.5:1.5b

in the ollama,

Reply with only one Linux command to show network interfaces.

It runs



3.2.1.3 Created the directory and python environment:

Created the directory network-ai-agent

mkdir -p ~/network-ai-agent

entered the directory

cd ~/network-ai-agent

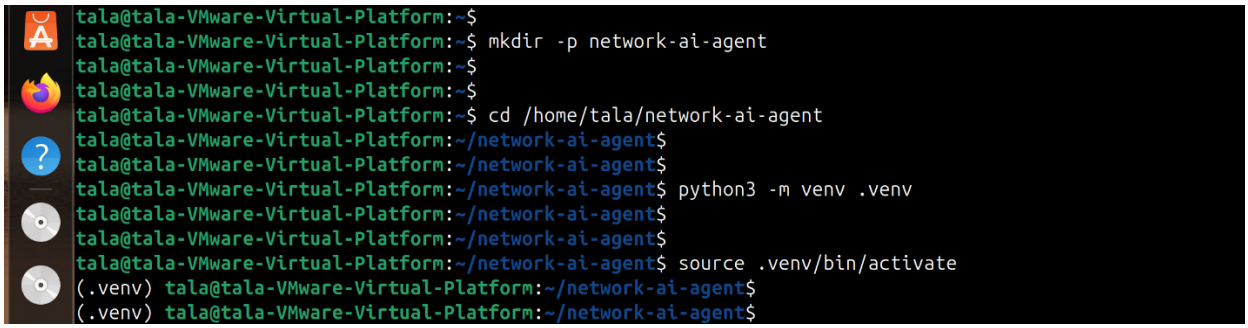
Created the virtual environment

python3 -m venv .venv

enable the environment

source .venv/bin/activate

success, displayed (.venv)

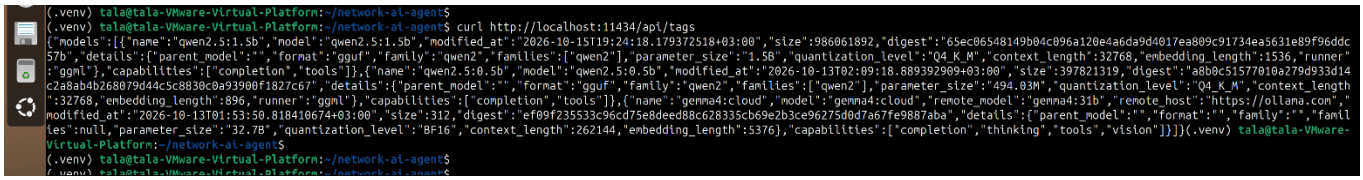


3.2.1.4 Testing:

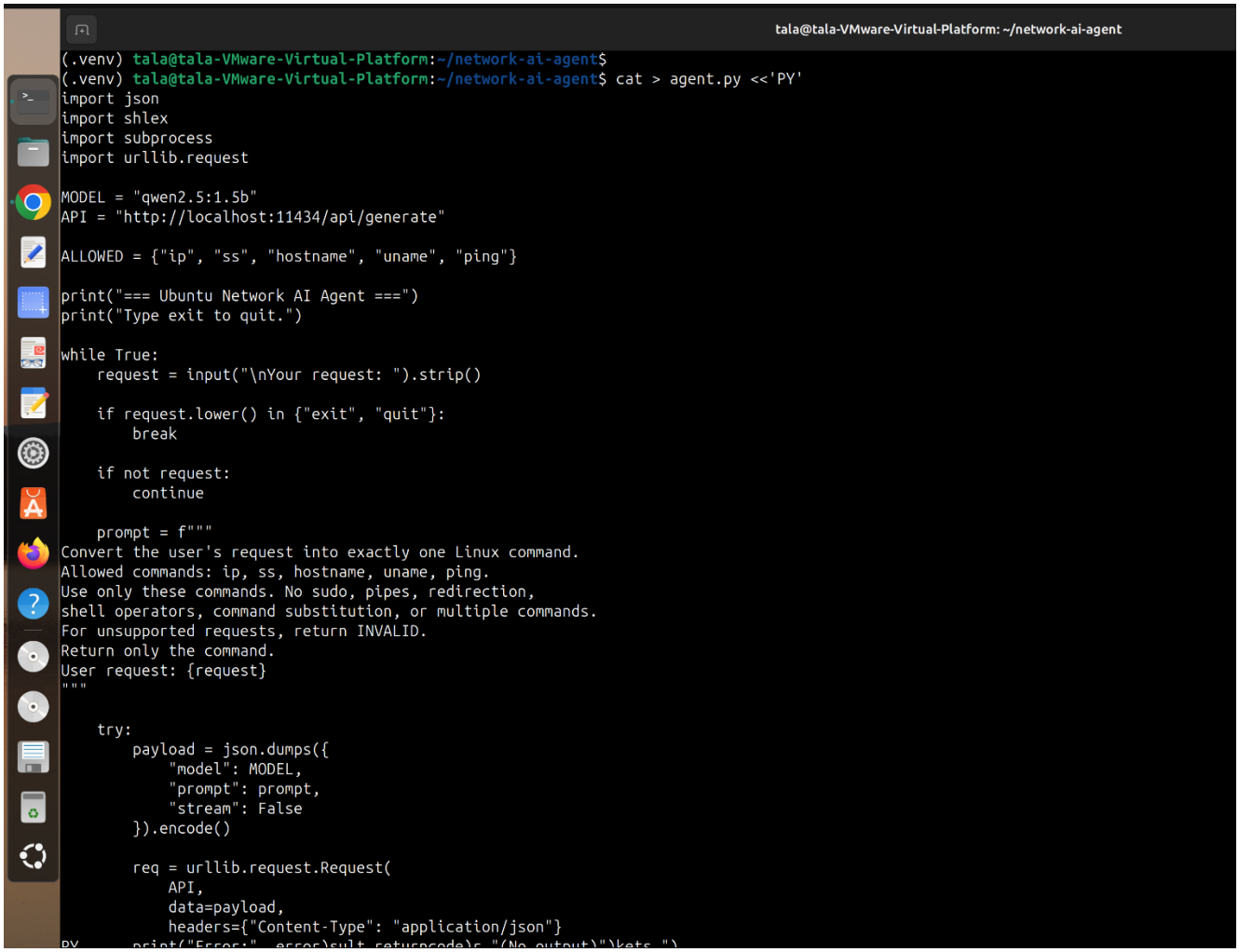
Checked that Ollama is responding

curl http://localhost:11434/api/tags

responded with JSON that contains models



3.2.1.5 Created the python file:



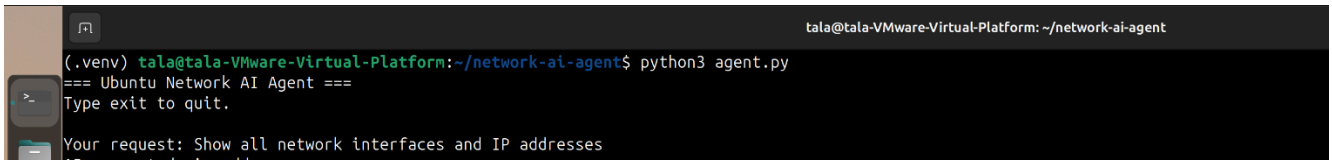
The agent also includes command restrictions and an approval step before executing an action.

The agent generated the requested actions and displayed the actual results when execution succeeded.

3.2.1.6 Implementation:

Run the AI agent

python3 agent.py

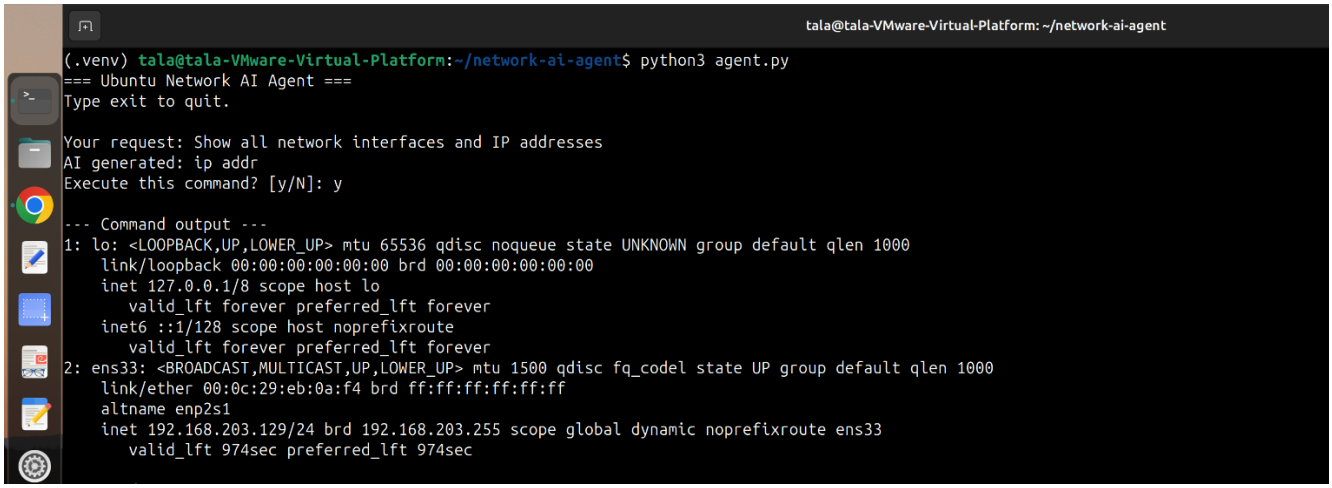


1. Display network interfaces and IP addresses.

Show all network interfaces and IP addresses

The suggestion command **(ip add)**

Approved the request of executing the command **(Y)**

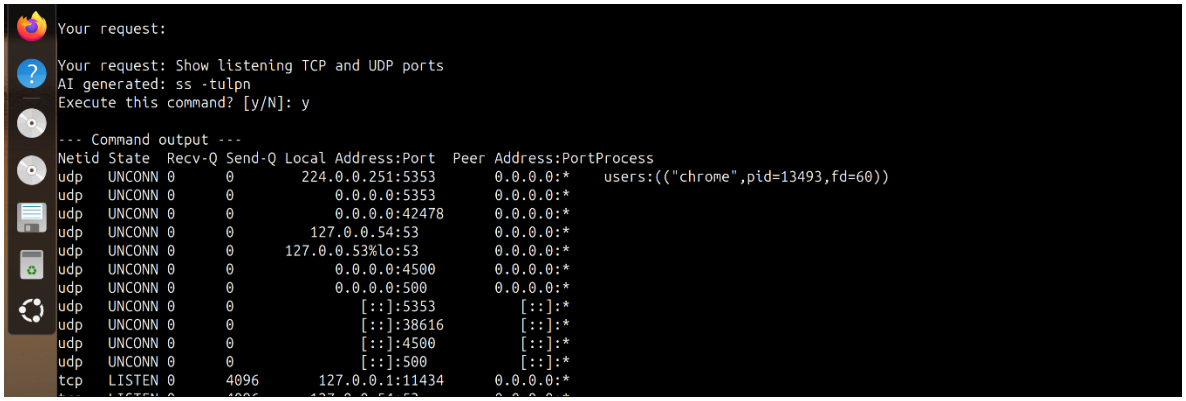


2. Display listening network ports.

Show listening TCP and UDP ports

The suggestion command **(ss -tulpn)**

Approved the request of executing the command **(Y)**



3. Retrieve the system hostname.

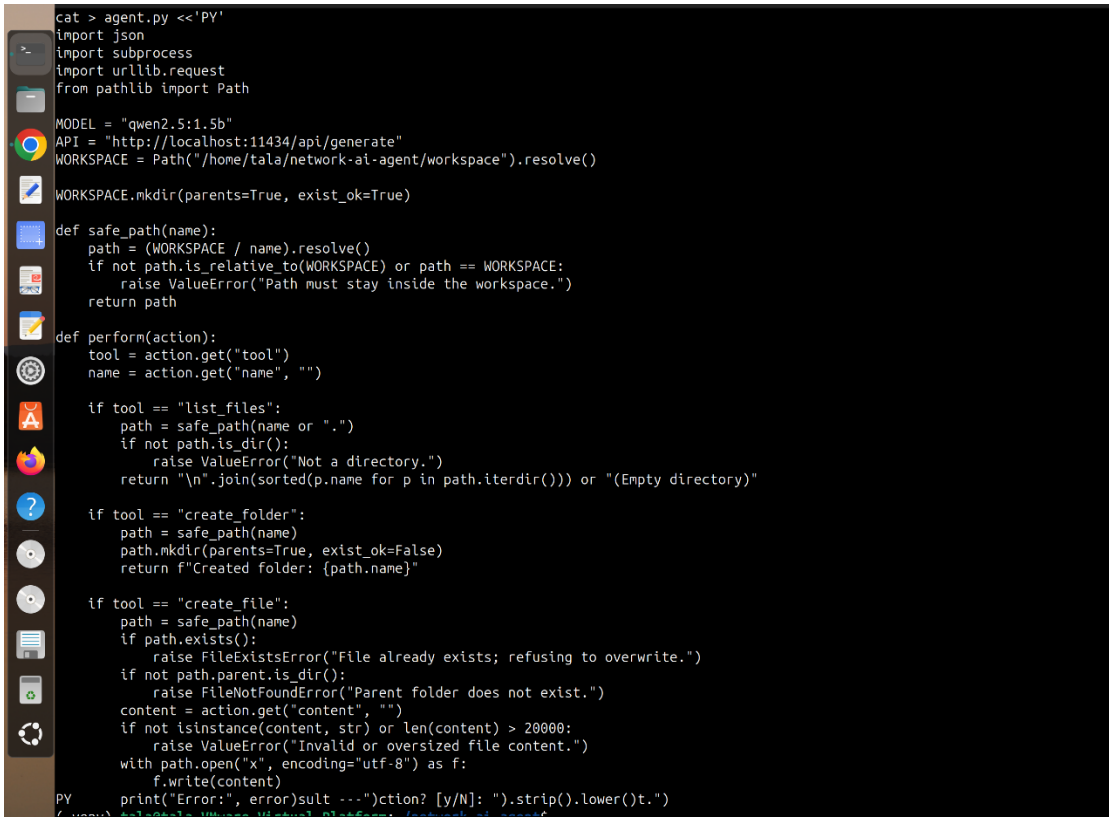
Show my computer hostname

The suggestion command **(hostname)**

Approved the request of executing the command **(Y)**



Developed the python file to execute more commands.



```
cat > agent.py <<'PY'
import json
import subprocess
import urllib.request
from pathlib import Path

MODEL = "qwen2.5:1.5b"
API = "http://localhost:11434/api/generate"
WORKSPACE = Path("/home/tala/network-ai-agent/workspace").resolve()

WORKSPACE.mkdir(parents=True, exist_ok=True)

def safe_path(name):
    path = (WORKSPACE / name).resolve()
    if not path.is_relative_to(WORKSPACE) or path == WORKSPACE:
        raise ValueError("Path must stay inside the workspace.")
    return path

def perform(action):
    tool = action.get("tool")
    name = action.get("name", "")

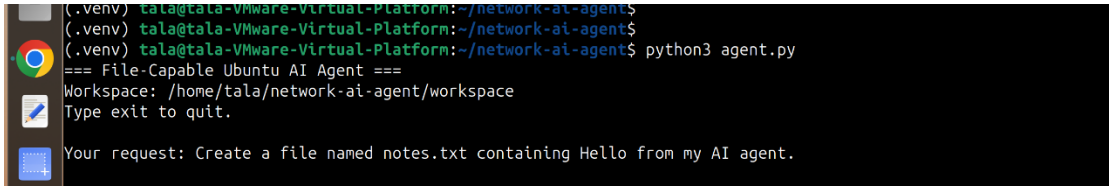
    if tool == "list_files":
        path = safe_path(name or ".")
        if not path.is_dir():
            raise ValueError("Not a directory.")
        return "\n".join(sorted(p.name for p in path.iterdir())) or "(Empty directory)"

    if tool == "create_folder":
        path = safe_path(name)
        path.mkdir(parents=True, exist_ok=False)
        return f"Created folder: {path.name}"

    if tool == "create_file":
        path = safe_path(name)
        if path.exists():
            raise FileExistsError("File already exists; refusing to overwrite.")
        if not path.parent.is_dir():
            raise FileNotFoundError("Parent folder does not exist.")
        content = action.get("content", "")
        if not isinstance(content, str) or len(content) > 20000:
            raise ValueError("Invalid or oversized file content.")
        with path.open("x", encoding="utf-8") as f:
            f.write(content)
PY      print("Error:", error)sult ---")ction? [y/N]: ").strip().lower()t.")
```

Run the AI agent again

python3 agent.py



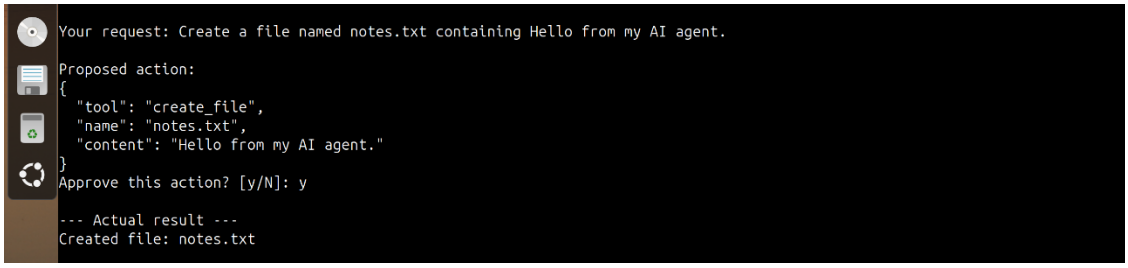
```
(.venv) tala@tala-VMware-Virtual-Platform:~/network-ai-agent$
(.venv) tala@tala-VMware-Virtual-Platform:~/network-ai-agent$
(.venv) tala@tala-VMware-Virtual-Platform:~/network-ai-agent$ python3 agent.py
=== File-Capable Ubuntu AI Agent ===
Workspace: /home/tala/network-ai-agent/workspace
Type exit to quit.

Your request: Create a file named notes.txt containing Hello from my AI agent.
```

4. Create text files and directories.

Create a file named notes.txt containing Hello from my AI agent.

Approved the request of executing the command **(Y)**



5. Read the contents of the file notes.txt.

Show notes.txt

Approved the request of executing the command **(Y)**



3.2.2 Method Two: AI Agent Using ShellGPT:

3.2.2.1 Tools Used

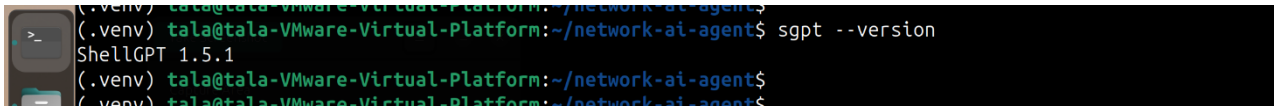
6. Ollama
7. Qwen 2.5:1.5B language model
8. ShellGPT 1.5.1

3.2.2.2 Installation and Configuration:

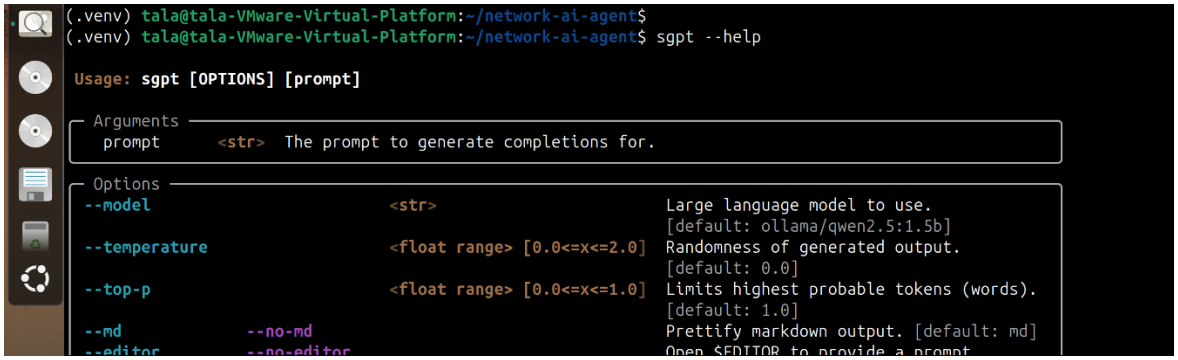
Configured ShellGPT to use the local Ollama model.

Checked its version and available options using:

sgpt –version



sgpt --help



run these commands:

To know the sgpt file location, it was installed previously

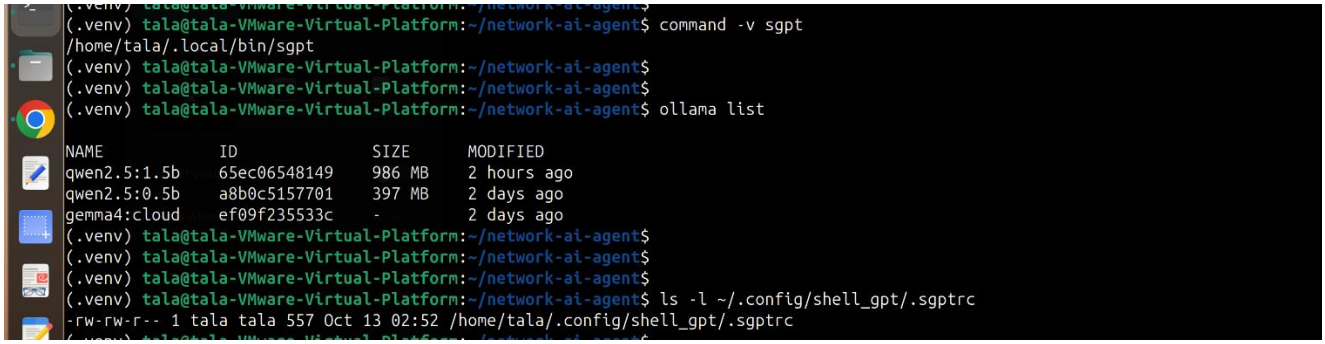
command -v sgpt

To display the ollama models

ollama list

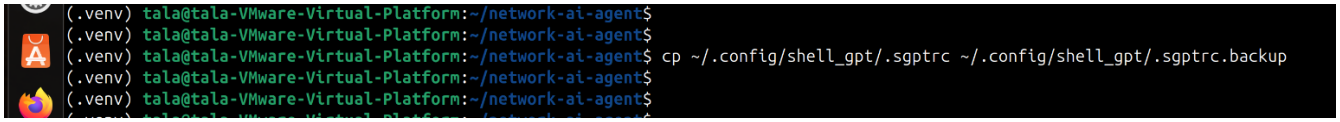
To examine the ShellGPT configuration file.

ls -l ~/.config/shell_gpt/.sgptrc



The settings file exists, made a backup of it

cp ~/.config/shell_gpt/.sgptrc ~/.config/shell_gpt/.sgptrc.backup



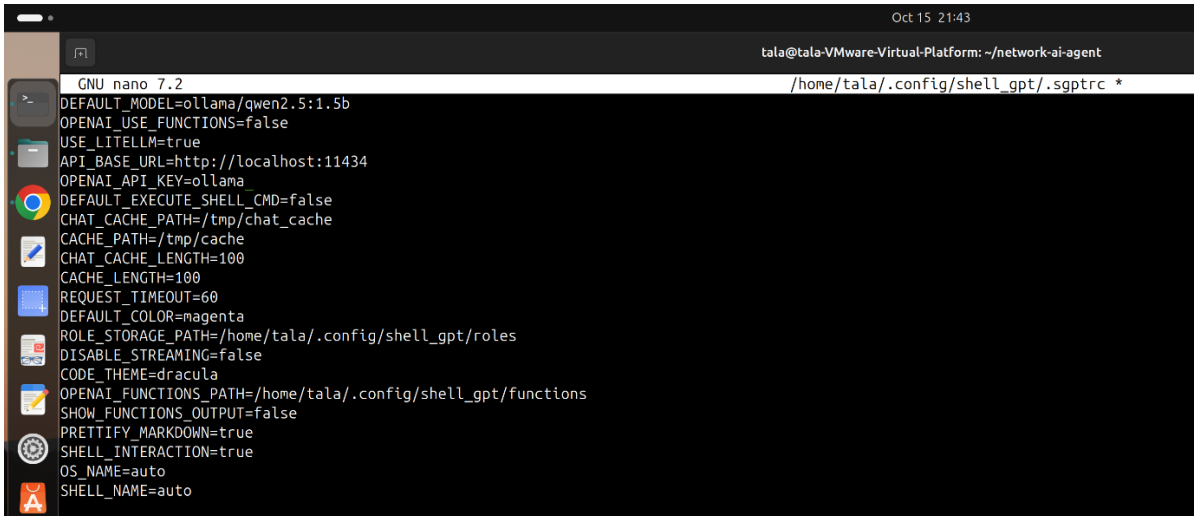
Opened it

nano ~/.config/shell_gpt/.sgptrc



Changed the settings, will set it up to use Ollama locally, instead of requiring an OpenAI API key. Save and exit

OPENAI_API_KEY=ollama



3.2.2.3 testing:

sgpt "Give me one Linux command to list files"

the answer is: **ls**



sgpt --shell "Show network interfaces and their IP addresses"

ShellGPT generated a command and displayed the following options:

Execute (E): Execute the generated command.

Modify (M): Modify the command.

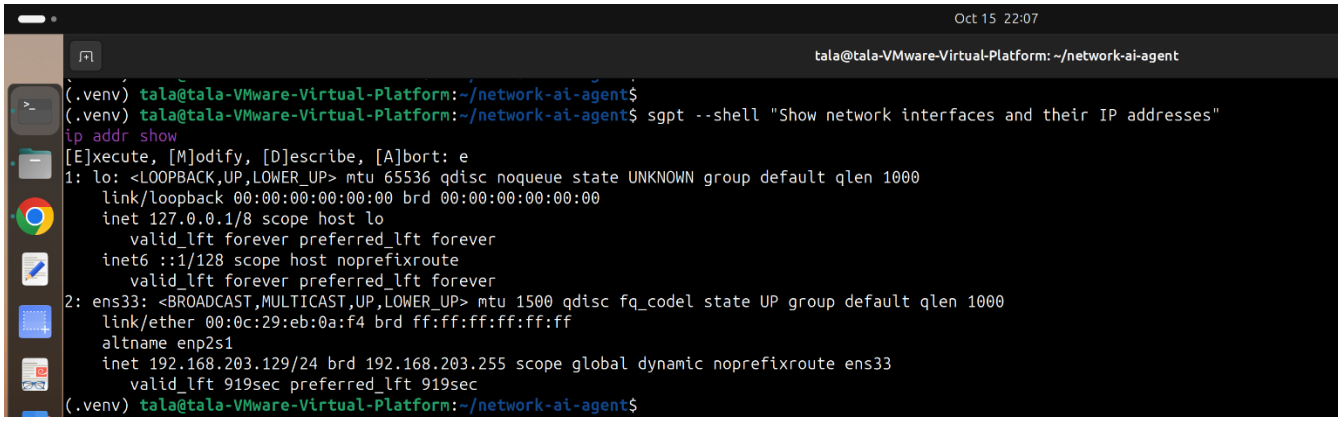
Describe (D): Explain the command.

Abort (A): Cancel execution.

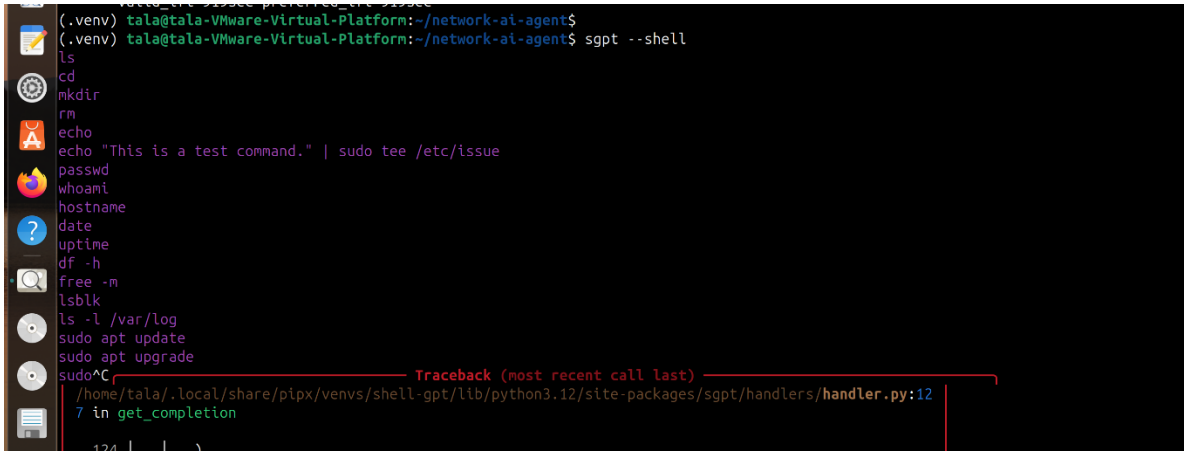
Selecting Execute ran the command and displayed its actual output.

the answer is: **ip addr show**

for more details, confirm execute (**e**)

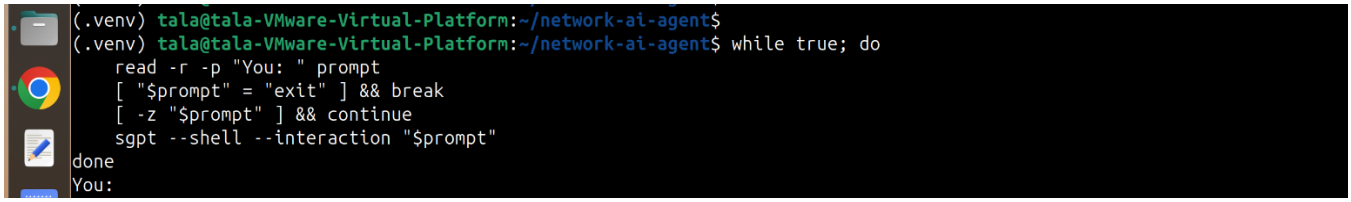


Sgpt --shell



3.2.2.4 Interactive Execution Loop:

To allow multiple requests without manually restarting ShellGPT each time, this loop repeatedly accepts requests, launches ShellGPT, and returns to the input prompt after each command interaction.



3.2.2.5 Implementation:

1. Displays network interfaces and their IP addresses.

Show network interfaces and their IP addresses

ShellGPT generated: **ip addr show**

Then wrote (**e**) for more information

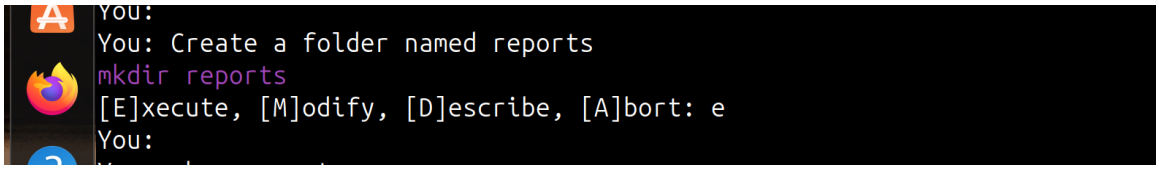
After execution was approved, Ubuntu displayed the available network interfaces and their assigned IP addresses.



2. Create a folder with name reports.

Create a folder named reports

ShellGPT generated: **mkdir reports**

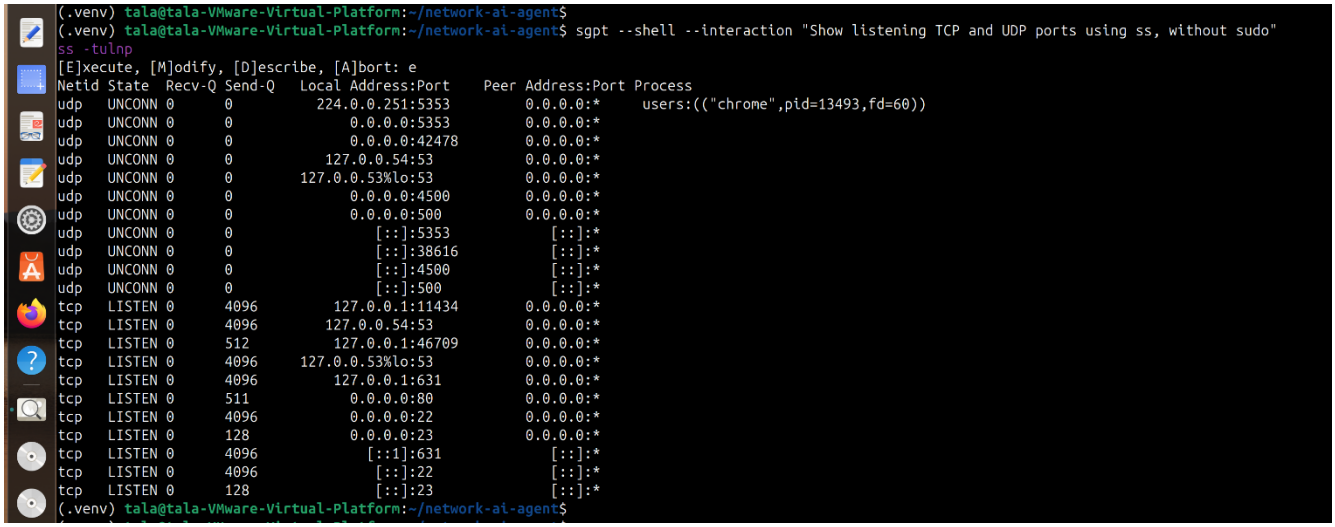


3. Display the TCP, UDP listening ports.

sgpt --shell --interaction "Show listening TCP and UDP ports using ss, without sudo"

ShellGPT generated: **ss -tulnp**

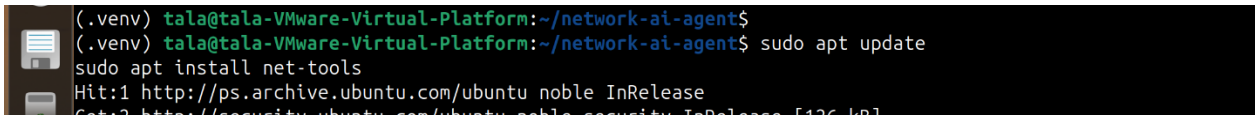
Then wrote (**e**) for more information



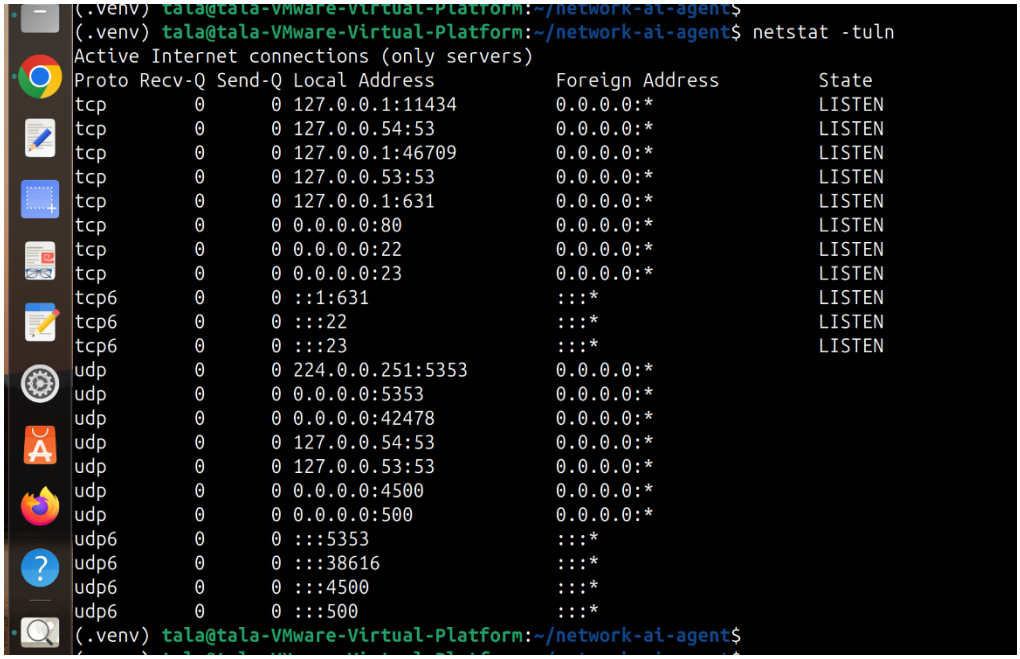
Executed the commands, to use the netstat tool

sudo apt update

sudo apt install net-tools



netstat -tuln

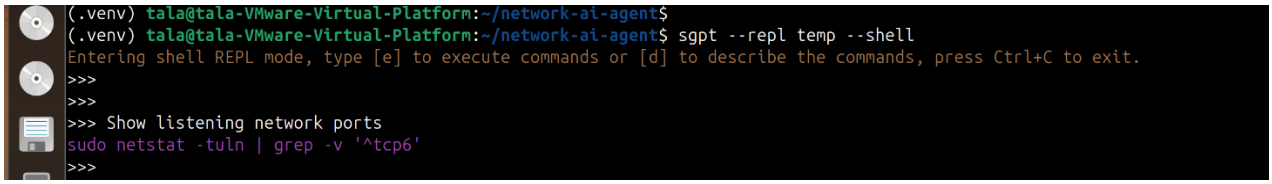


4. Show listening network ports.

sgpt --repl temp --shell

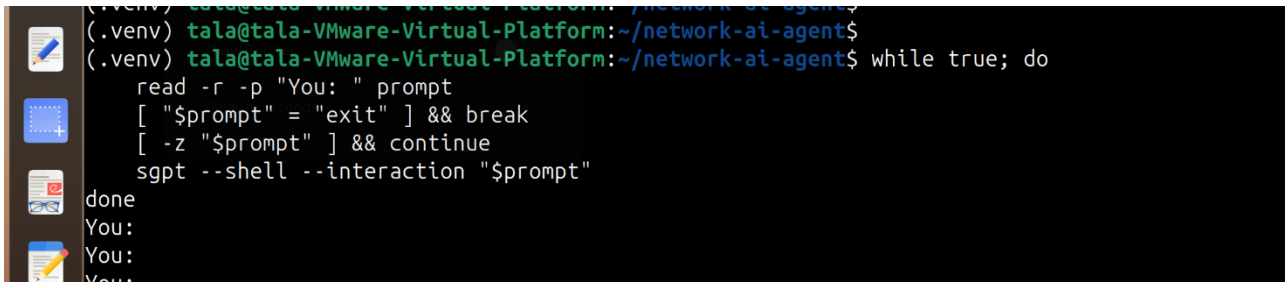
Show listening network ports

ShellGPT generated: sudo netstat -tuln | grep -v ‘^tcp6’



To continue in the same session, on terminal, run

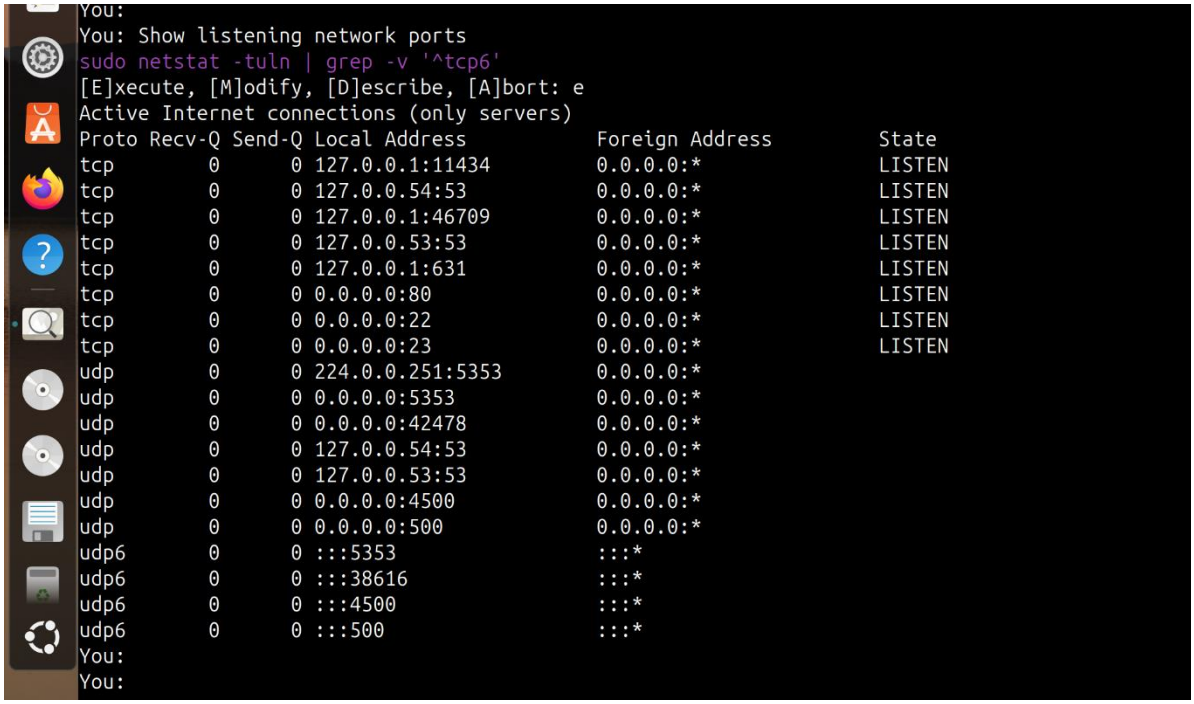
```
while true; do
    read -r -p "You: " prompt
    [ "$prompt" = "exit" ] && break
    [ -z "$prompt" ] && continue
    sgpt --shell --interaction "$prompt"
done
```



Show listening network ports

ShellGPT generated: sudo netstat -tuln | grep -v ‘^tcp6’

Then wrote (**e**) for more



4. Conclusion:

4.1 commands:

Ppracticing Linux commands helped understand how to manage files and directories, view and process text, and obtain system information using the terminal. These commands are essential for basic Linux system administration.

4.2 AI agent:

The AI agent demonstrated how artificial intelligence can help generate and execute Linux commands, such as checking network interfaces and listening ports. This makes system administration easier and more efficient while still requiring the user to review commands before execution.